

CHAPTER

3

Maintenance Adjustment and Replacement

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3.1 Mechanical Unit Configuration

■ SS type



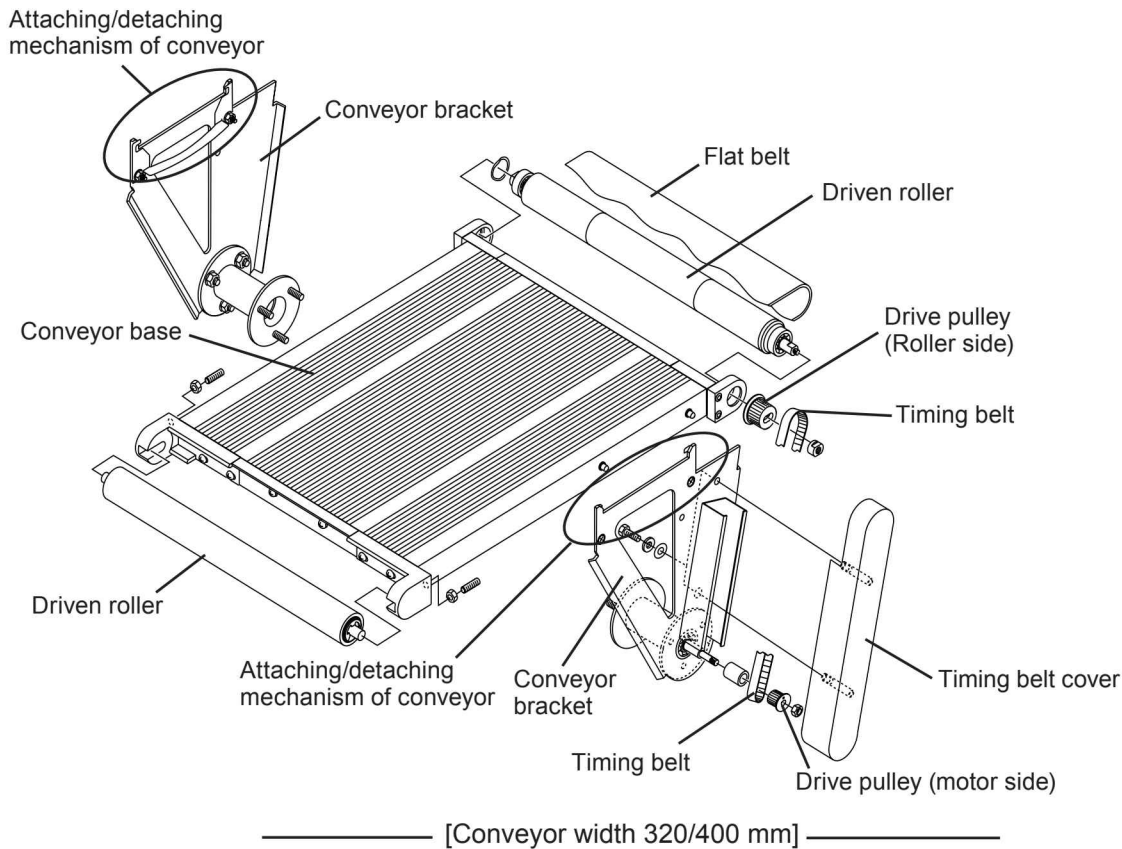
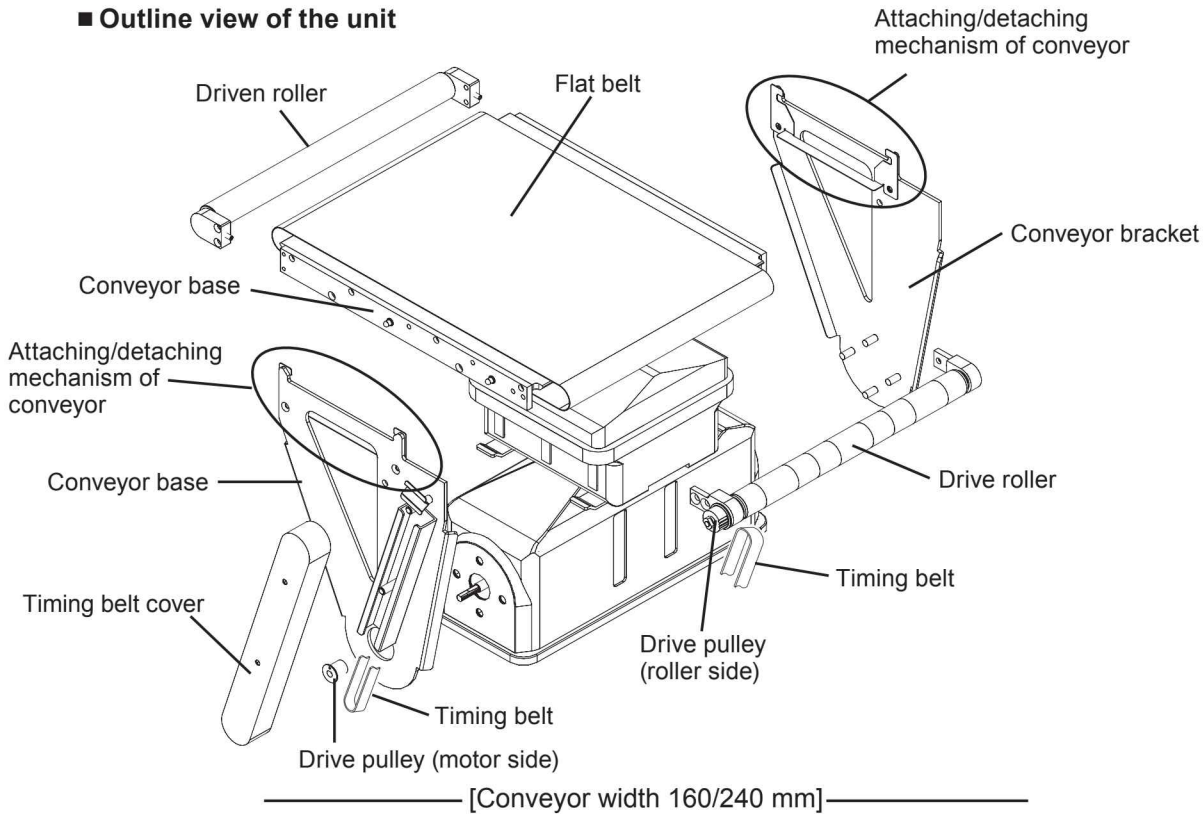
■ CR type



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3.2 Infeed Conveyor (SS Type)

■ Outline view of the unit



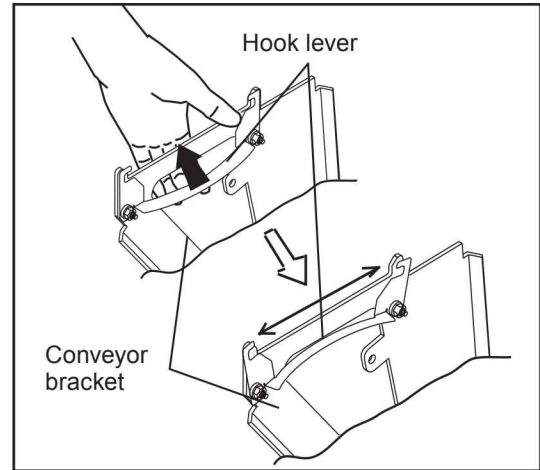
■ Unit function

- Carrying products to weigh conveyor.

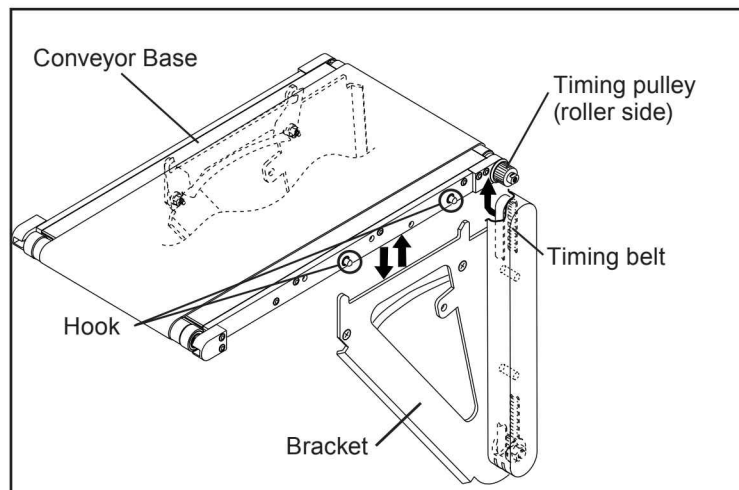
3.2.1 How to detach and attach infeed conveyor

In general, the infeed conveyor is detached and attached at the time of cleaning after operation or during maintenance such as belt replacement. Though it can be easily detached/attached by means of the one-touch detach/attach mechanism, how to detach and attach it is described below for confirmation.

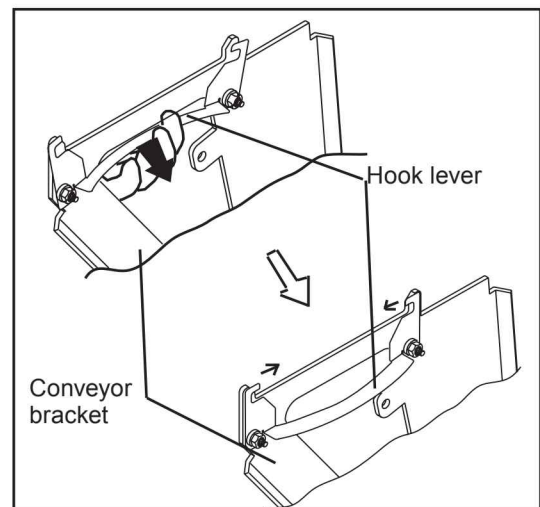
1. Lightly lift up the plate spring of the conveyor bracket with the fingers as shown on the right figure. Then the lever hooks are opened to the right and left. The conveyor base can be now detached. (It can be easily detached by means of the one-touch detach/attach mechanism.)



NOTE When detaching the driving pulley (roller side), remove the conveyor by pulling out the driving pulley from the timing belt.



2. Place the conveyor base hook on the securing section of the conveyor bracket, and lightly pull down the plate spring with the fingers as shown on the right figure. Then the lever hooks move in the central direction. The conveyor base can be now secured. (It can be easily attached by means of the one touch detach/attach mechanism.)



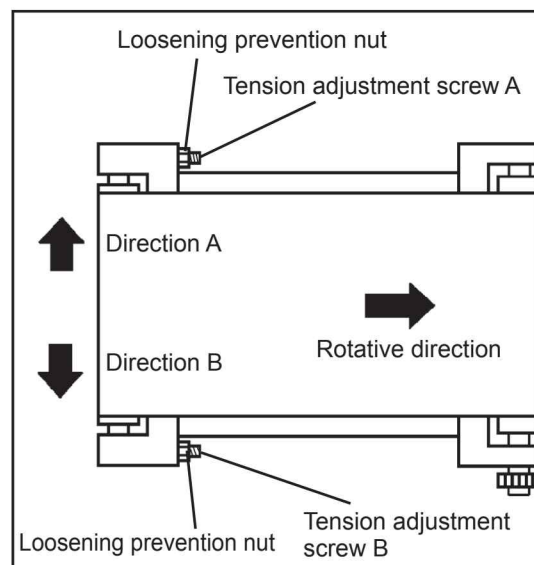
NOTE Secure the conveyor base to the conveyor bracket after hanging the driving pulley (roller side) on the timing belt.

3.2.2 Tension and meandering adjustments of the infeed conveyor belt

The tension of the flat belt for the infeed conveyor is adjusted with the tension adjustment crews.

Perform the following steps to adjust the tension of the flat belt. When adjusting the meandering of the flat belt, start from step 3.

1. Make sure that the infeed conveyor is stopped, and tighten tension adjustment screws A and B so that the tension of the flat belt (transfer belt) becomes uniform at both sides to the extent that the driving roller and flat belt (transfer belt) do not run idle.
2. Then tighten the screws by one more turn.



NOTE Make sure that the tension of the flat belt (transfer belt) is uniform at both sides.

3. Set the high belt speed reservation. Touch the operation  key on the main menu of the operation panel.
 - The infeed conveyor starts to operate.
4. Check the inclination direction of the flat belt (transfer belt).
5. When the flat belt (transfer belt) inclines in the A direction, tighten tension adjustment screw A or loosen tension adjustment screw B. When the flat belt (transfer belt) inclines in the B direction tighten tension adjustment screw B or loosen tension adjustment screw A.
6. When inclination of the flat belt (transfer belt) has been removed, lock tension adjustment screws A and B with the loosening prevention nuts. Then, touch the operation  key
 - The infeed conveyor stops operation.



Be careful not to tighten tension adjustment screws A and B too much. The tensile condition in step 2 is optimal and the tension reaches the limit if the screws are tightened by a half more turn.

(If the screws are tightened too much, durability of the bearings extremely becomes worse.)

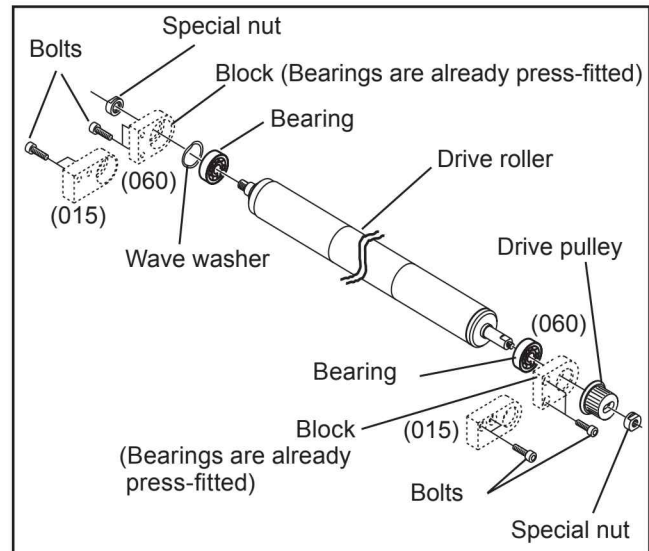
CAUTION

- If the tension is not appropriate, the flat belt (transfer belt) motion may become unstable.
- If the tension is not appropriate, the motor may stop due to overload.
- If the tension is not appropriate, the weighing accuracy may become worse.
- If the conveyor encounters a temperature gradient or sudden temperature change (for example, air conditioner's air directly blowing on the flat belt (transfer belt)), the flat belt (transfer belt) motion may become unstable.
- If objects like a broken bag get caught between the conveyor base and the flat belt (transfer belt), the flat belt (transfer belt) motion may become unstable.
- If an offset load is imposed on the flat belt surface of the conveyor (for example, rubbing of the guide), the flat belt (transfer belt) motion may become unstable.

3.2.3 Replacement of bearings for driving roller of infeed conveyor

Perform the following steps to replace bearings for the driving roller of the infeed conveyor.

1. Remove the screws for securing the blocks at both ends of the driving roller (2 screws for each block).
2. Remove the special nuts at both ends of the driving roller shaft.
3. Remove the driving pulley.
4. Remove the blocks at both ends.



5. Be careful not to lose a wave washer which is contained in one of the blocks. Be sure to insert the waved washer into the block.
6. Old bearings may be difficult to remove. In such cases, remove them so that the driving roller shaft is not damaged or deformed.
7. Attach new bearings and assemble the driving roller of the infeed conveyor in a retrograde order of step 1 to 5.

NOTE

When replacing only bearing, check if the bearing is not rattled due to the inside of the block is worn away. When attaching new bearings, do not apply any LOCTITE agent on the bearings.

3 Maintenance Adjustment and Replacement

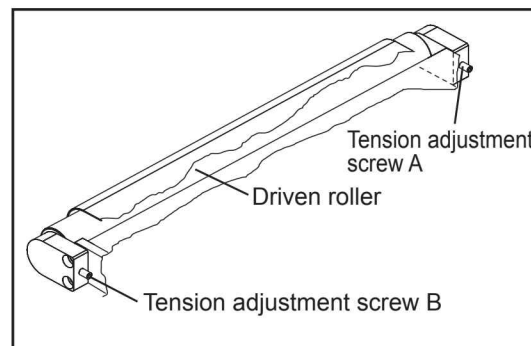
3.2.4 Replacement of driven roller for infeed conveyor

Perform the following steps to replace the driven roller and shaft for the infeed conveyor.

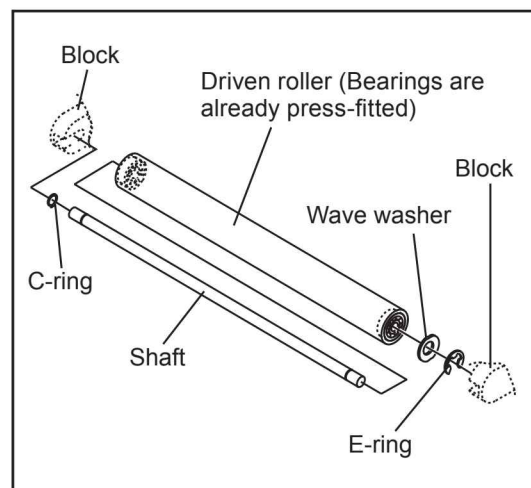
1. Loosen tension adjustment screws A and B and remove the flat belt (transfer belt).

NOTE

Carefully remove the flat belt (transfer belt), because the driven roller sometimes drops out of the block.



2. Remove the C ring of the driven roller shaft, and then the shaft.
3. Attach the shaft removed in step 2 to a new driven roller.



4. Attach the C ring to the driven roller shaft. Attach the driven roller unit to the blocks, and then attach the flat belt (transfer belt). Tighten tension adjustment screws A and B and perform "3.2.2 Tension and meandering adjustments of the infeed conveyor belt".

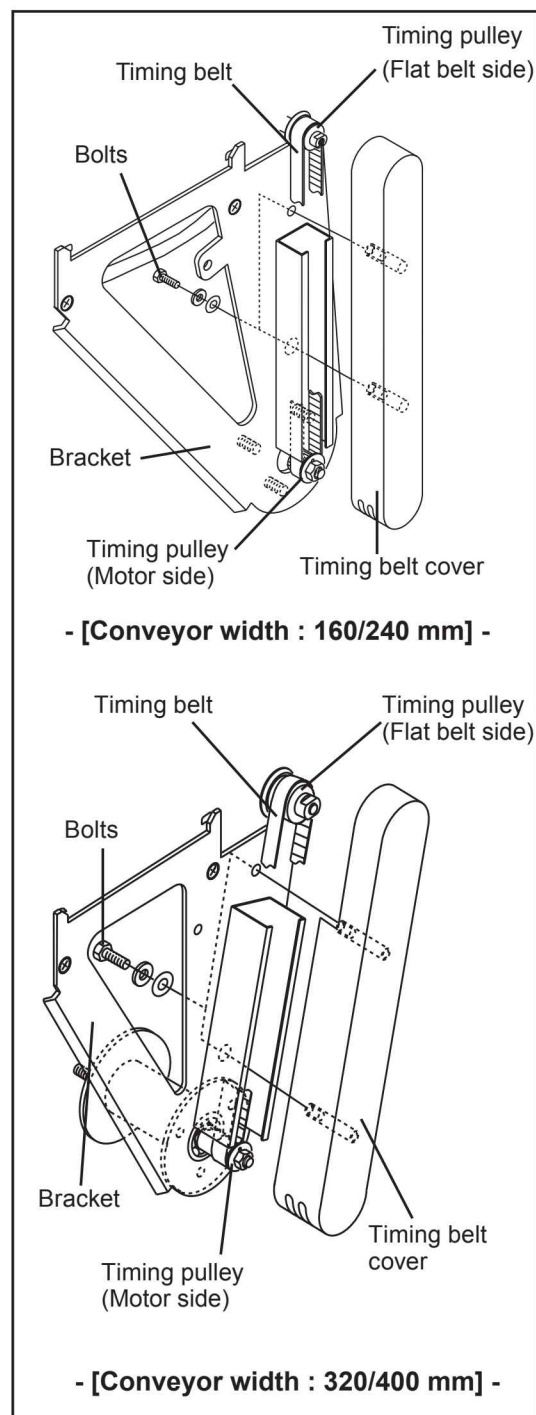
NOTE

Do not perform an on-site replacement of only bearings for the driven roller unit of the infeed conveyor. Replace the driven roller unit. (Because bearings are press-fitted in the driven roller.)

3.2.5 Replacement of timing belt

Perform the following steps to exchange the timing belt.

1. Remove 2 securing screws for the timing belt cover inside the conveyor bracket, and then remove the timing belt cover.
2. Remove the old timing belt.
3. Attach a new timing belt to the driving pulley (flat belt side). At this time, slightly pull down the timing belt and hang it on the driving pulley (motor side).
4. Attach the timing belt cover to the conveyor bracket and secure them with 2 screws.

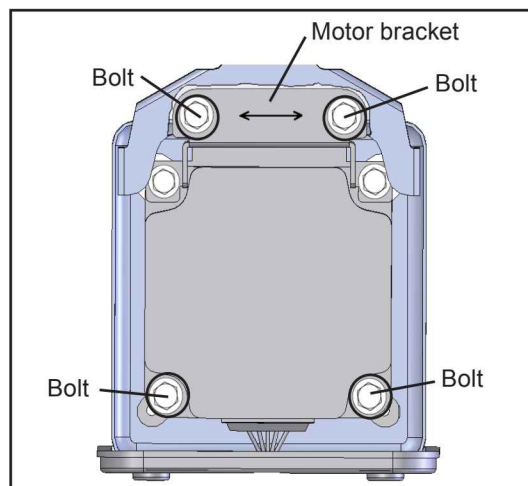


3.2.6 Adjusting tension of timing belt

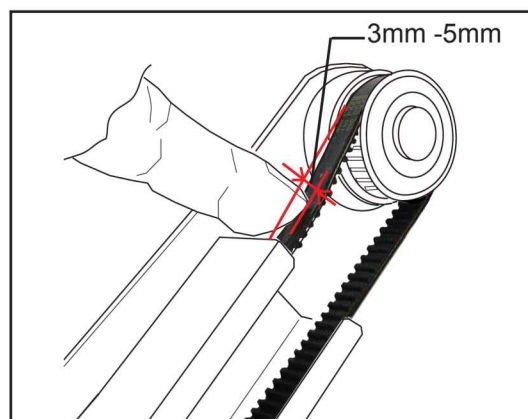
After replacing the timing belt and driving motor, check and adjust the tension of timing belt by the following steps.

■ For the conveyor of 160/240 mm wide (common to SS, CR)

1. Slightly loosen four bolts fixing the motor bracket, move the motor bracket, and adjust the tension of the timing belt.
2. Tighten four bolts with the timing belt tightened.

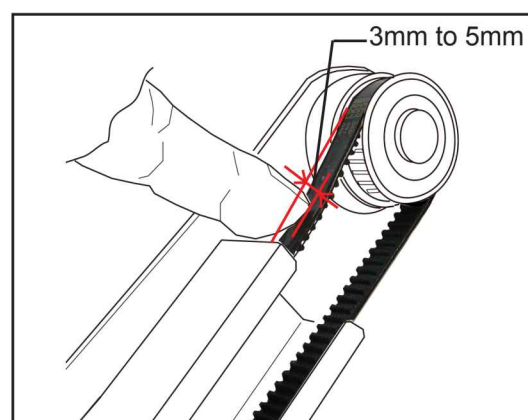
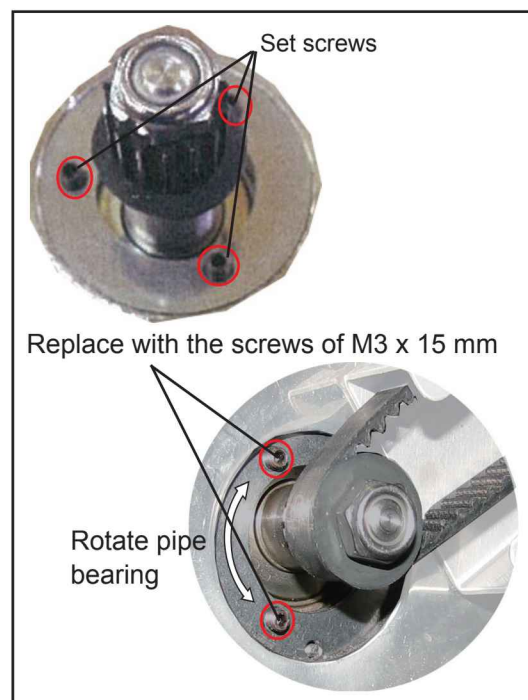


3. Gently press the timing belt at the position shown in the right figure. Check that the timing belt moves approximately 3 mm to 5 mm. If it does not move or moves less/more than the above range, adjust the tension again by referring to the step 1.



■ For the conveyor of 320/400 mm wide (common to SS, CR)

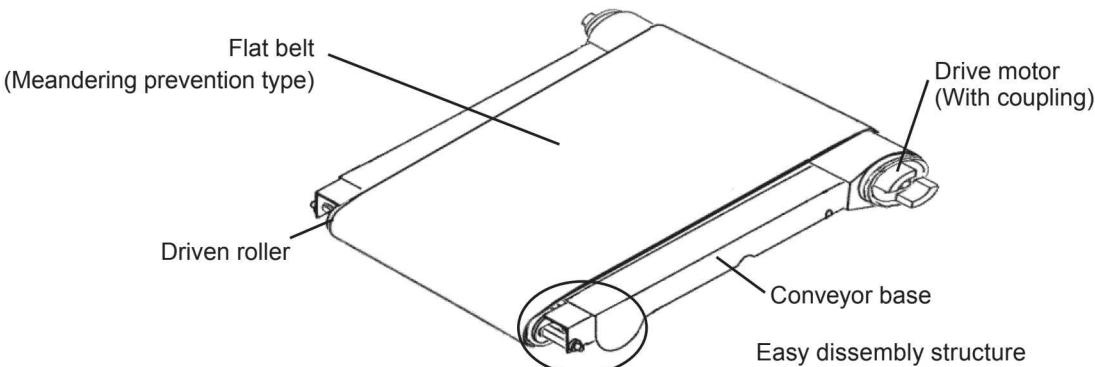
1. Loosen three set screws fixing the pipe bearing. Remove two set screws, and attach two screws of M3 x 15 mm.
2. Rotate the pipe bearing by using two screws of M3 x 15 mm, to adjust the tension of timing belt.
3. With the timing belt tightened, tighten the set screw. Remove two screws of M3 x 15 mm, attach the set screw, and tighten it.
4. Slightly press the timing belt at the position shown in the right figure. Check that the timing belt moves approximately 3 mm to 5 mm. If it does not move or moves less/more than the above range, adjust the tension again by referring to the steps 1 to 3.



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3.3 Infeed Conveyor (CR Type)

■ Outline view of the unit



■ Unit function

Carrying products to weigh conveyor



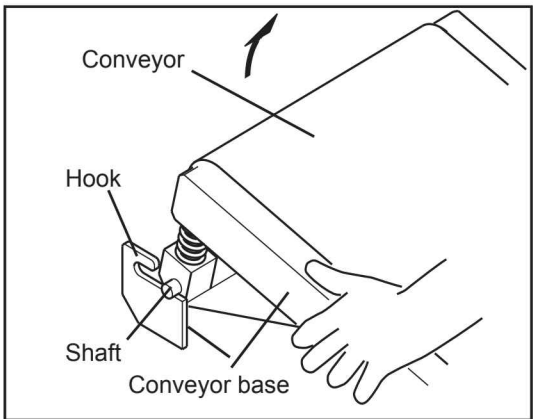
- Since the conveyor belt of CR specification is a meandering prevention type, the meandering adjustment of the standard conveyor belt is not necessary.
- If assembling the conveyor unit of CR specification on site, adjust the length of the conveyor on both sides by means of the one-touch assembly/disassembly mechanism on the driven roller side in order to prevent it from meandering

3.3.1 Disassembly of Infeed Conveyor

Usually, infeed conveyor is disassembled for cleaning after operation or maintenance such as replacement of belts. Easy disassembly structure enables you to disassemble the conveyor as shown below.

(Metal detector integral type infeed conveyor can be also disassembled by easy disassembly structure. Be careful for tension rollers at 2 places.)

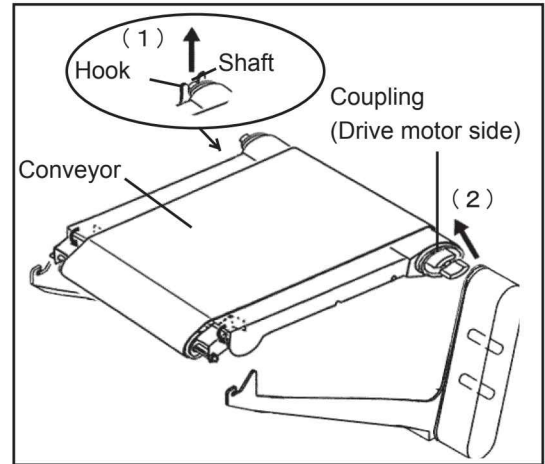
1. Lift up the intake side of infeed conveyor to release the shaft from the hook as shown in the right figure. (Removal can be easily done by the easy disassembly structure.)



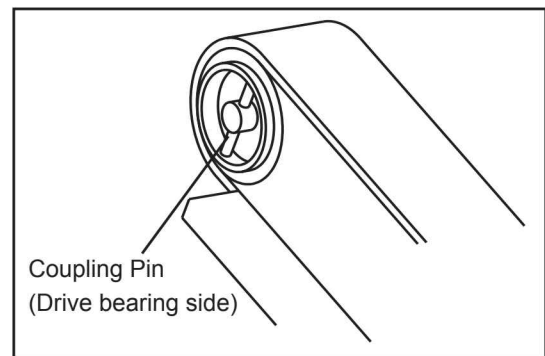
2. As in the right figure,
 - 1) Lift the discharged side of infeed conveyor to the direction of arrow mark.
 - 2) Lift up the conveyor (drive roller) by pulling it to the direction of arrow mark, and remove the conveyor.

NOTE

Be careful not to damage the coupling of the drive roller because it may be a little hard to disassemble it.



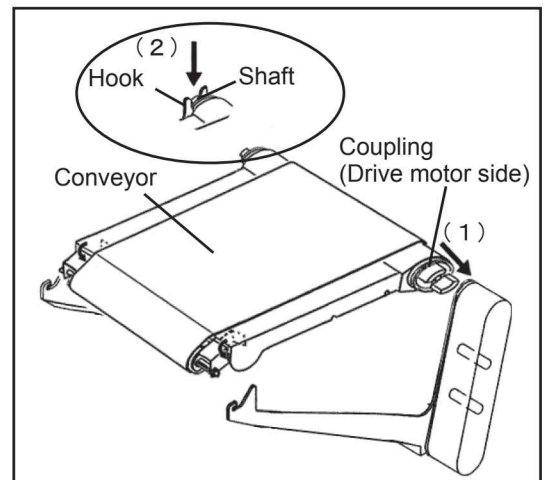
3. Remove the infeed conveyor, clean and check it, then assemble it again.
4. Align the coupling notch of the drive roller side of the infeed conveyor with the coupling pin of the drive bearing side.



NOTE

In assembling the drive roller coupling, ensure correct setting; the coupling pin may be hard to fit the notch of weigh conveyor.

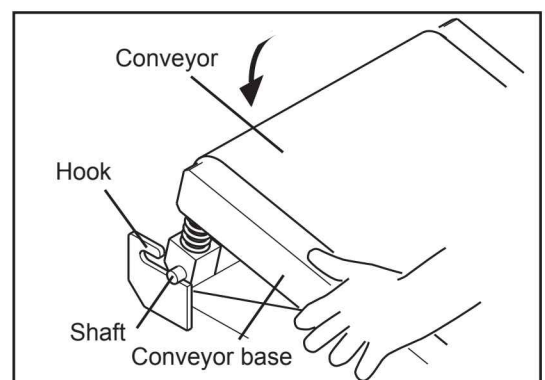
5. Insert the conveyor (drive roller) at the discharged side of the infeed conveyor to the direction (1), then to the direction (2).



NOTE

Be sure to hold the conveyor base by your fingers and then push the conveyor base so as not to get your finger caught.

6. Hold the intake side of the infeed conveyor by gripping the conveyor base as shown in the right figure, and press the conveyor. (Easy disassembly structure helps you to attach it easily.)

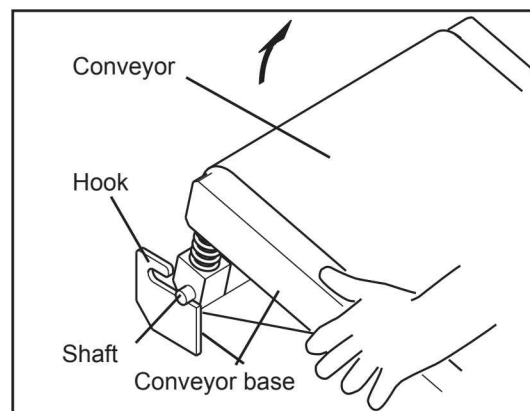


3.3.2 Replacing Infeed Conveyor Belt

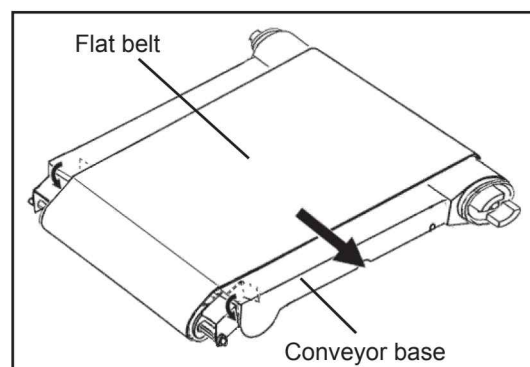
Usually, infeed conveyor is disassembled for cleaning after operation or maintenance such as replacement of belts. Easy disassembly structure enables you to disassemble the conveyor as shown below.

1. Remove infeed conveyor by the procedure shown in "3.3.1 Disassembly of Infeed Conveyor". (It can be disassembled by the easy disassembly structure.)

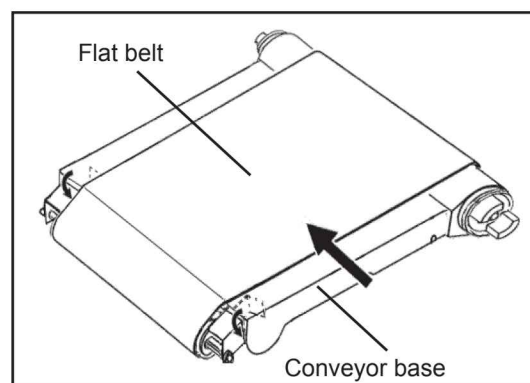
NOTE It may be a little hard to disassemble the coupling of drive roller; take care.



2. From the disassembled infeed conveyor, remove the flat belt to the direction shown in the right figure. (Easy disassembly structure helps you to attach it easily)

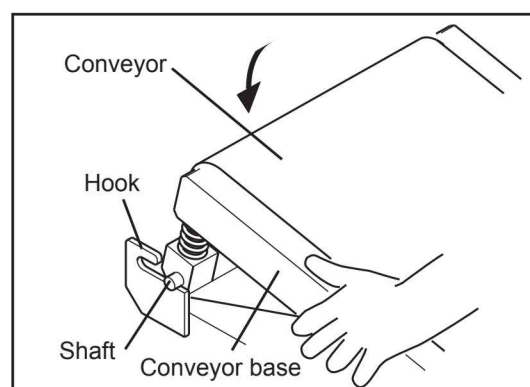


3. Install new flat belt to the infeed conveyor base to the direction as shown in the right figure.



4. Attach the infeed conveyor by the same procedure shown in "3.3.1 Disassembly of Infeed Conveyor". (Easy disassembly structure helps you to attach it easily.)

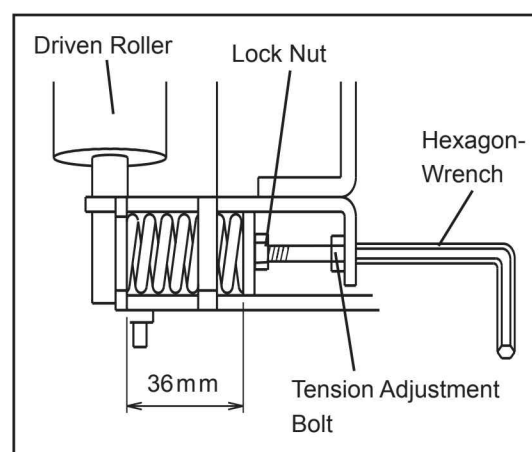
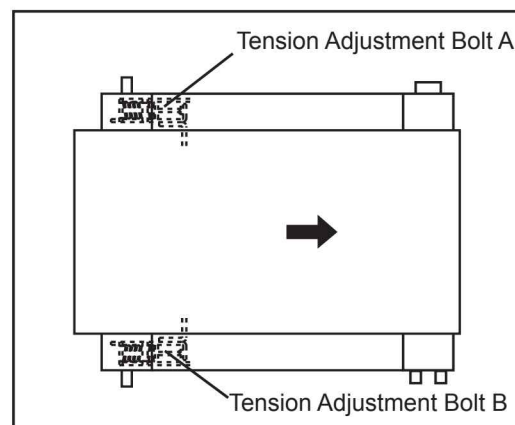
NOTE When pushing the shaft at the conveyor end in the hook, hold the conveyor base with a finger push it in carefully taking care not to be pinched.



3.3.3 Tension and meandering (snaking) adjustments of infeed weigh conveyors

The tension of the flat belt for infeed conveyor must be adjusted by the adjustment bolt. Follow the instruction below for the tension adjustment of the flat belt. In case of adjustment of the lateral shifting for the flat belt, follow the instruction from 2.

1. Adjust the spring length to 36 mm by adjustment bolt at the driven roller side of infeed conveyor.
2. Press  key to make infeed conveyor active. Check and see if flat belt is in the proper (center) position. Then, press  key to complete the tension adjustment.
3. Turn the tension adjusting bolts A/B equally to adjust the tension of the conveyor belt.
4. Make sure that the conveyor guide fit the roller's groove and no excessive force is applied.



NOTE

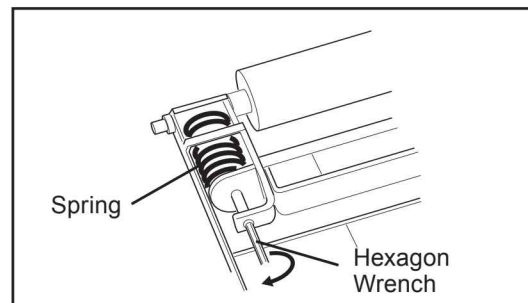
- Since the conveyor belt of CR specification is a meandering prevention type, the meandering adjustment of the standard conveyor belt is not necessary.
- Still, if the right/left tension adjustment is poor, the meandering prevention guide of flat belt becomes worn to cause powder swarf.
- Adjust the length of the conveyor on both sides by means of the one-touch assembly/disassembly mechanism on the driven roller side in order to prevent it from meandering.

3 Maintenance Adjustment and Replacement

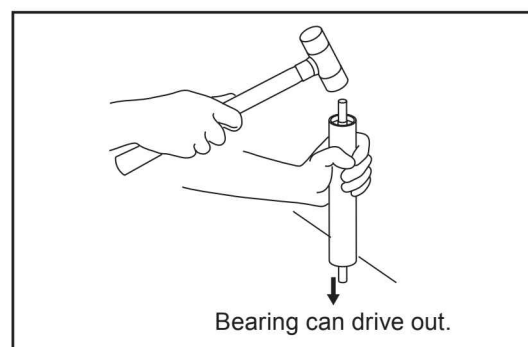
3.3.4 Replacement of driven roller bearing (and oil seal) for conveyor

For Replacement of driven roller bearing (and oil seal) for conveyor, follow the poucire.

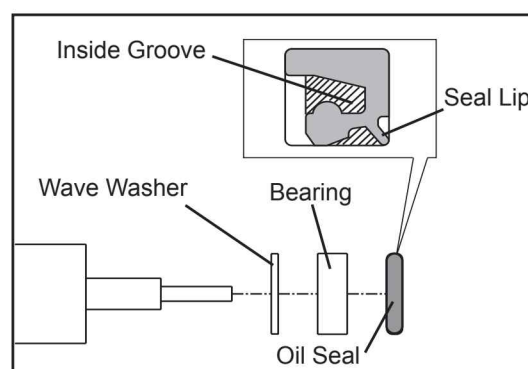
1. Unfasten (rotate clockwise) the set screw by hexagon wrench to decrease the spring force and remove the spring that fixes the both ends of driven roller.
2. Remove the driven roller, and remove the oil seal from the roller shaft.
3. Tap the roller shaft with a plastic hammer to drive out the bearing at one end.



NOTE Do not excessively hit the bearing. Otherwise, it will be damaged.



4. Drive out the other bearing from the roller using a round bar.
5. With the reverse procedure to above, assemble the wave washer, the bearing and the oil seal, and attach the driven roller to the conveyor.



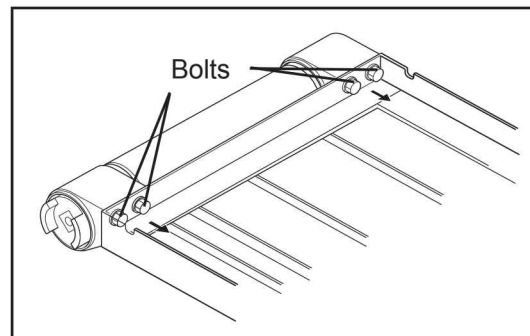
NOTE

- When mounting the oil seal, sufficiently apply the specified grease to the inside grove and the seal lip. (The shaded area must be greased.)
- Specified grease: Kluber Lubrication, PARALIQ (R) GTE703

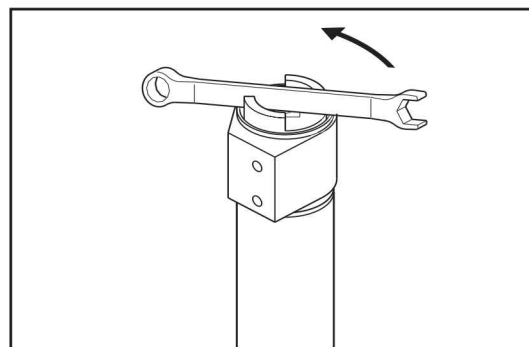
3.3.5 Replacement of belt-side bearing (and oil seal) for conveyor drive roller

For replacement of belt-side bearing (and oil seal) for conveyor drive roller, follow the procedures below:

1. Take out 4 bearing fixing bolts and remove the bearing of drive roller.



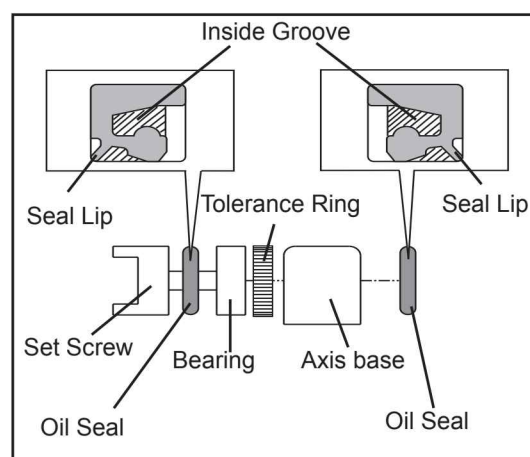
2. Rotate the set screw of the bearing at coupling side counterclockwise as shown at right, and remove the bearing.



3. Insert a screwdriver into the shaft hole to take 2 oil seals out of the bearing.

4. Strike out the bearing by a round bar to remove the bearing.

5. By the reverse procedure to above, assemble a bearing and an oil seal, and attach the bearing to the conveyor.



NOTE

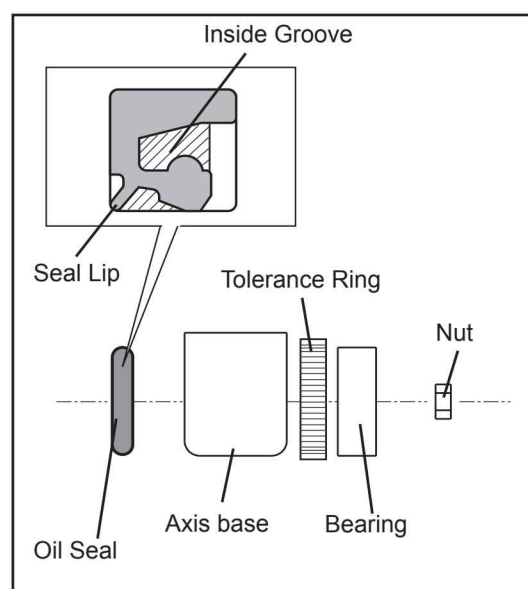
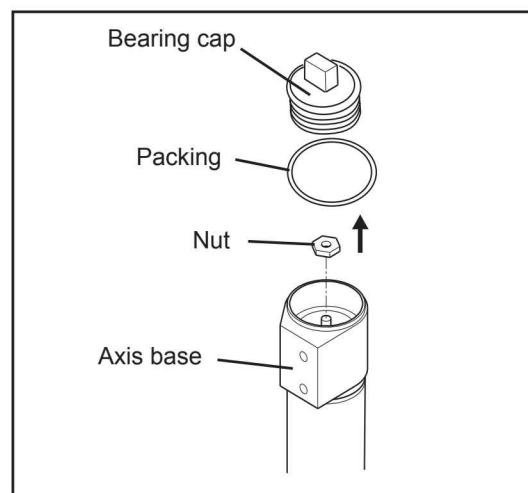
- When mounting the oil seal, sufficiently apply the specified grease to the inside groove and the seal lip. (The shaded area must be greased.)
- Specified grease: Kluber Lubrication, PARALIQ (R) GTE703

3 Maintenance Adjustment and Replacement

3.3.6 Replacing the Bearing (Oil Seal) of Conveyor Drive Roller at Anti-Coupling Side

Follow the instruction below for replacing the bearing (oil seal) of the conveyor drive roller at anticoupling side.

1. Take out 4 bearing fixing bolts and remove the bearing of drive roller.
2. Remove the bearing fixing the driven roller.
3. Remove the nut inside the bearing to remove the bearing.
4. Insert a screwdriver in the shaft hole and take 2 oil seals out of the bearing.
5. Strike out the bearing by a round bar to remove the bearing.
6. With reverse procedure to the above, assemble a bearing and an oil seal, and attach the bearing to the conveyor.

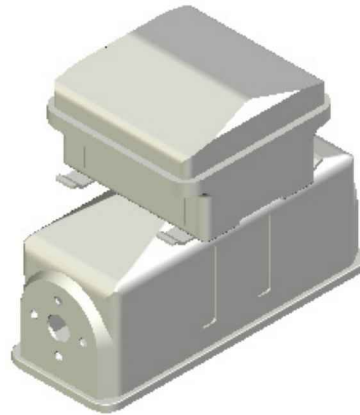


NOTE

- When mounting the oil seal, sufficiently apply the specified grease to the inside groove and the seal lip. (The shaded area must be greased.)
- Specified grease: Kluber Lubrication, PARALIQ (R) GTE703

3.4 Drive Section of Infeed Conveyor

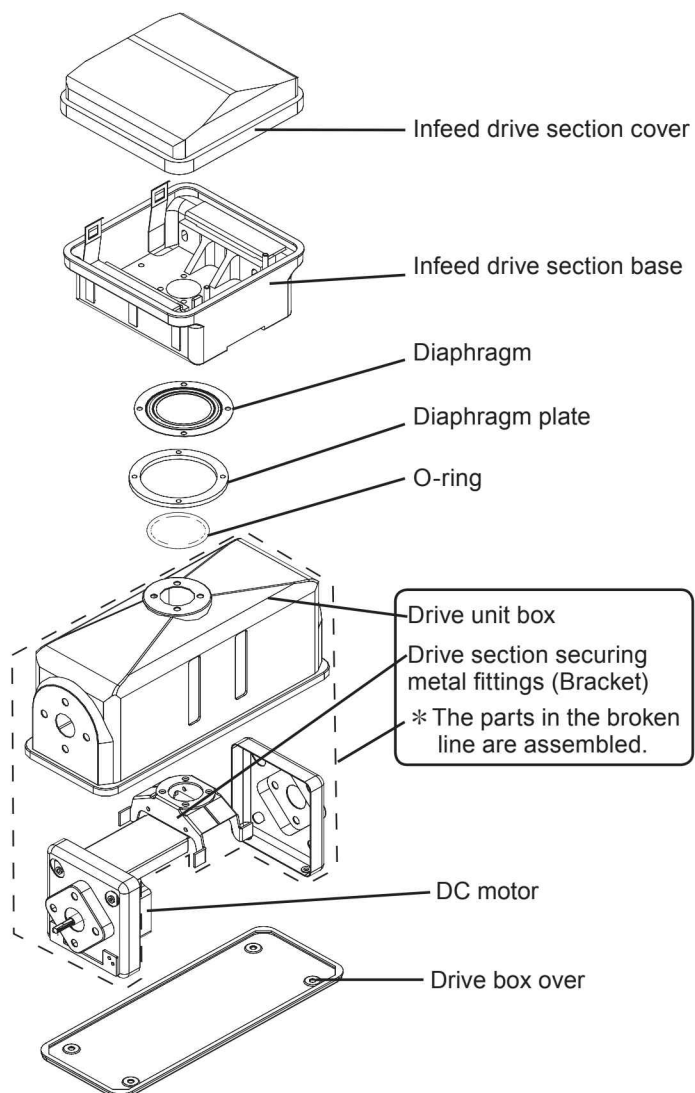
■ Outline view of the unit



■ Function of the unit

- It transmits the motor drive to the driving roller of the infeed conveyor.

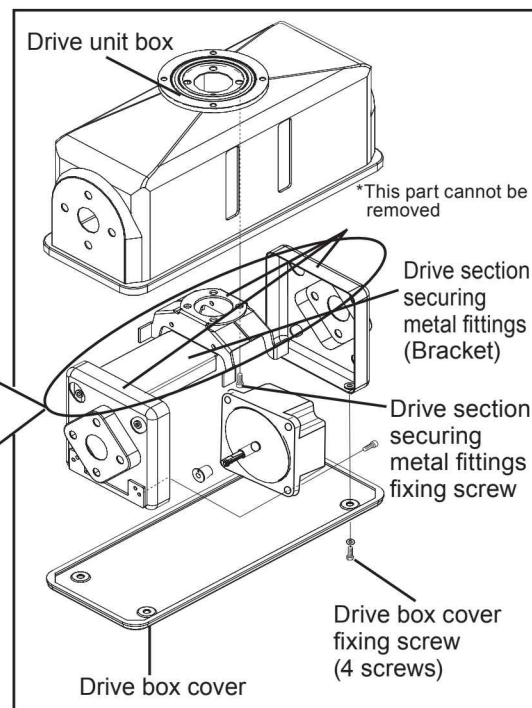
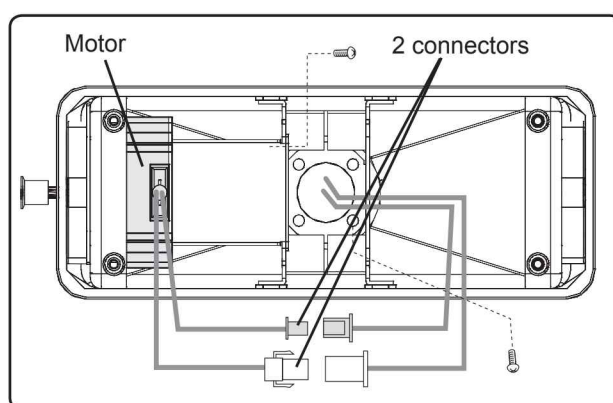
■ Exploded view of the unit



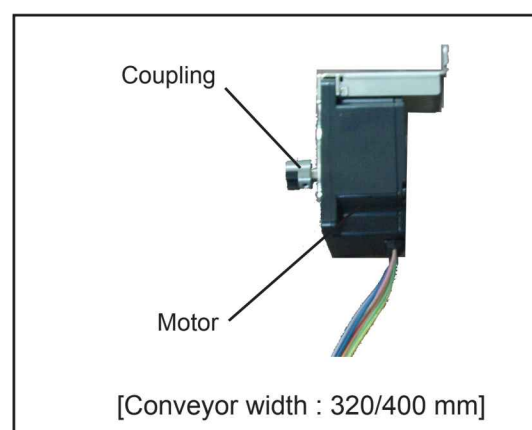
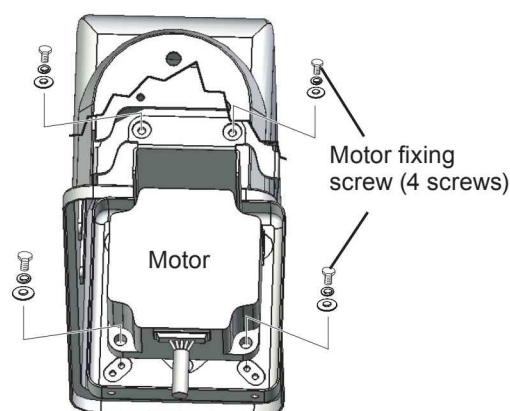
3.4.1 Replacement of drive motor

Perform the following steps to replace the drive motor of the infeed conveyor, drive shaft (015 type) bearings.

1. Remove the 4 drive box cover fixing screws at the bottom and remove the drive box cover.
2. Remove the 2 motor cable connectors.

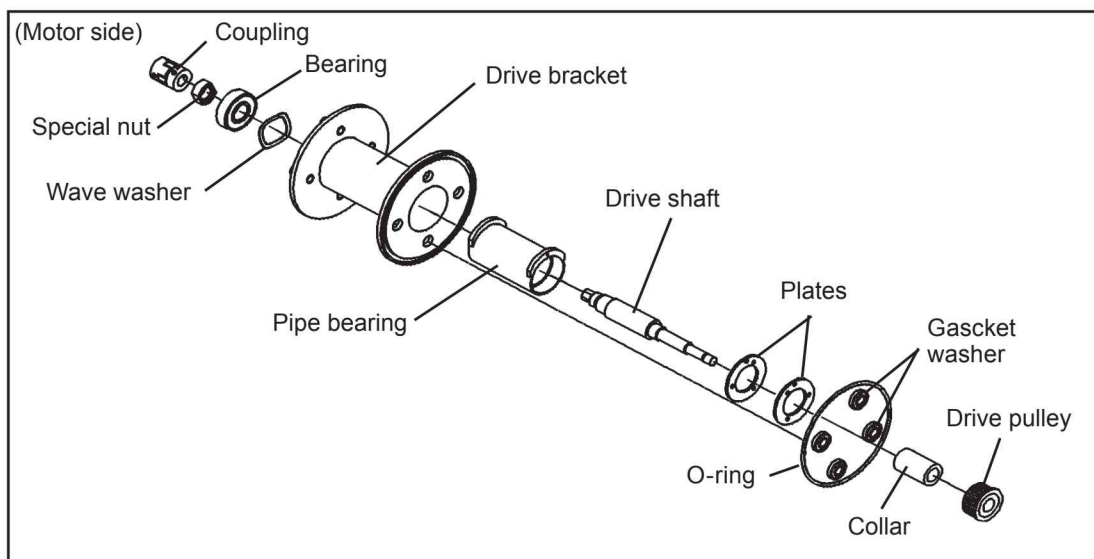


3. Remove 4 drive box securing screws and then remove the motor unit.
4. Regarding the DACS-G-060 machine, remove the timing belt cover of the conveyor bracket, and then loosen and remove the lock nut located on the drive shaft (motor side). Then remove the driving pulley (motor side).
5. Remove 4 motor securing screws and then remove the motor.
6. Attach a new motor and assemble the components in a retrograde order of step 1 to 4.



NOTE When replacing the motor of the conveyor width 320/400mm machine, note that a coupling is contained in the motor drive shaft.

7. After removing the motor, remove 4 nuts that secure the drive shaft unit inside the drive box, and then remove the drive shaft unit keeping the conveyor bracket attached.
8. Remove 4 nuts that secure the conveyor bracket, and then the conveyor bracket.



9. Loosen the special nut for the drive shaft unit as shown on the figure, and then remove the nut.

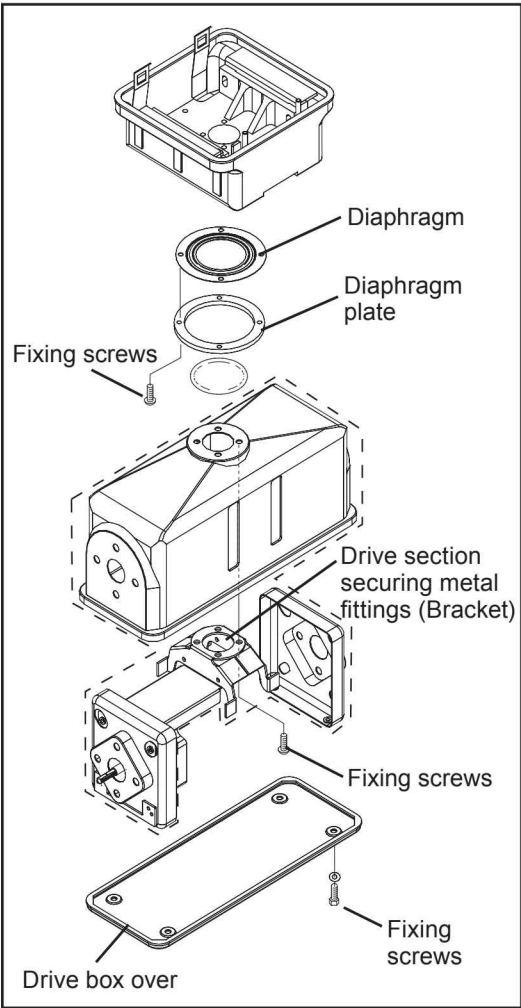
NOTE Be sure to set the special nut in the proper direction.

10. Remove the bearing (motor side).
11. Tap out the drive shaft to the pulley side with a plastic hammer and remove the drive shaft.
12. Remove the bearing (pulley side).
13. Set a new bearing (pulley side). At this time, insert a tolerance ring into the drive bracket and then press-fit a bearing.
14. Assemble the components in a retrograde order of step 7 to 12.

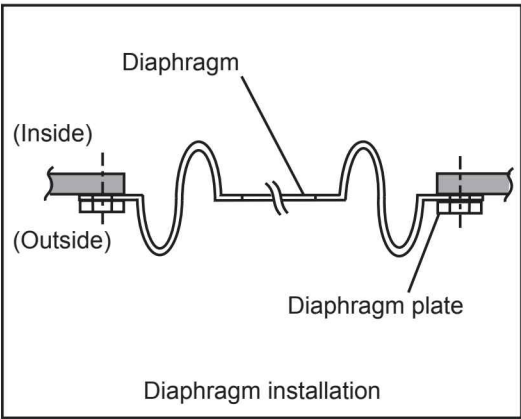
3.4.2 Replacement of diaphragm

Perform the following steps to replace the diaphragm for the drive section of the infeed conveyor.

1. Remove the drive motor box in a similar manner to the "3.4.1 Replacement of the drive motor".
2. As shown on the right figure, remove 4 securing nuts that secure the drive section securing metal fittings, and then remove the fittings.
3. Remove 4 securing screws that secure the diaphragm retainer, and then remove the retainer.
4. Attach a new diaphragm.
5. Assemble the components in a retrograde order of step 1 to 2.

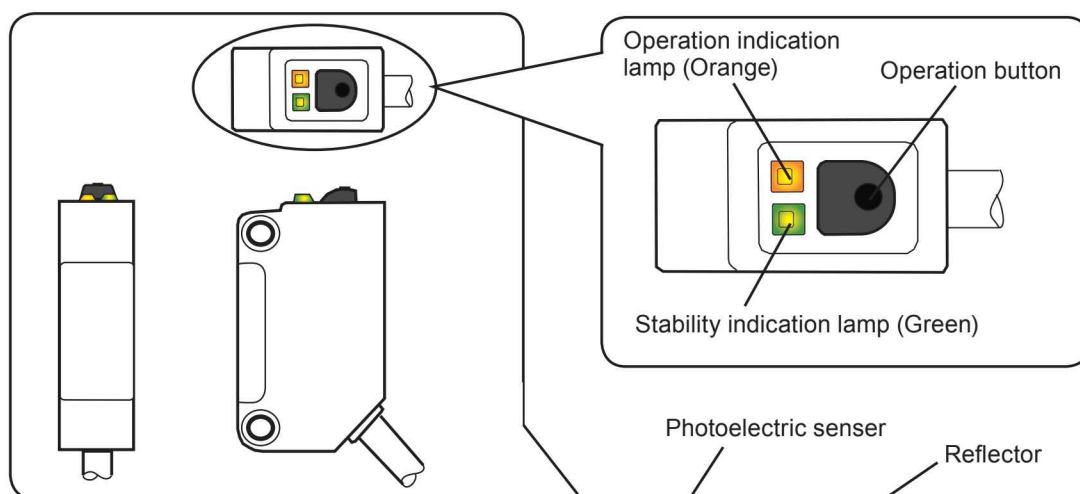


NOTE Attach the diaphragm as shown on the right figure.



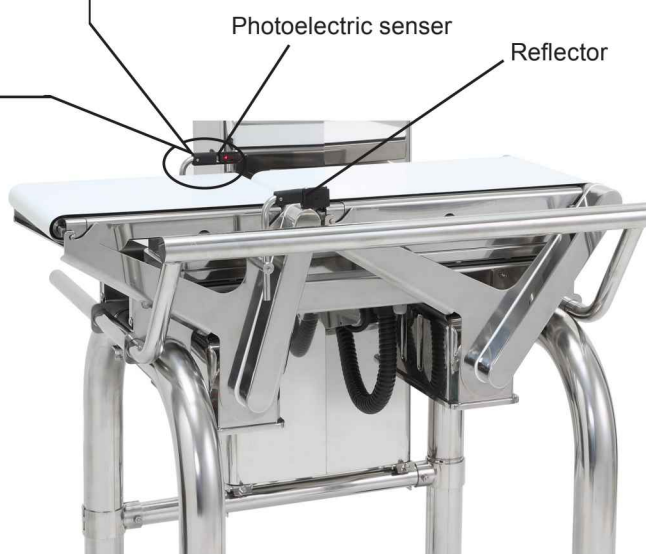
3.5 Photoelectric Unit

■ Outline view of the unit



■ Function of the unit

It senses that the product has been transferred to the weigh conveyor.



3.5.1 Photoelectric sensor and output stage circuit diagram

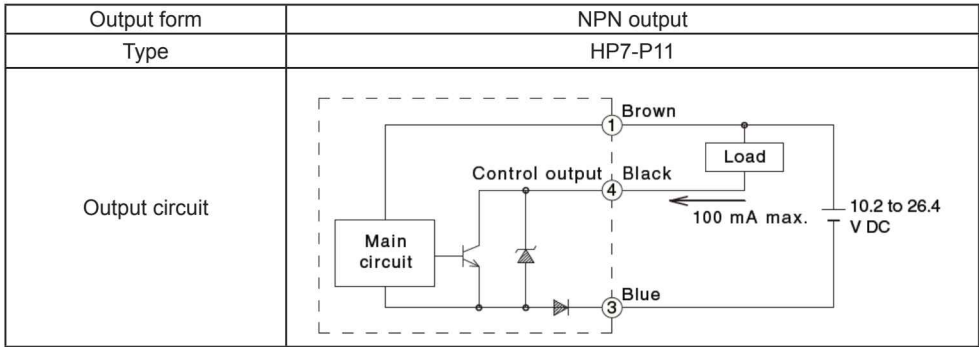
The rating and output stage circuit diagram of the photoelectric sensor (standard type) for this equipment [TYPE:HP7-P11] are shown below.

■ Specifications

Power supply	10.2 to 26.4 V DC
Power consumption	14 mA Max. (HP7-P11)
Control output	Open collector Switching current :Pre-wire/Pre-wire connector 100 mA (resistive load) Low - temperature resistance pre-wire/M8 connector 50 mA Voltage drop: 2 V Max. (at 100 mA switching current) 1.1 V Max. (Less than 10 mA)
Operating temperature and humidity	-30 to +55°C, 35 to 85% RH (without freezing or condensation) (except for low - temperature resistance pre-wire) Low - temperature resistance pre-wire : -30 to +40°C, 35 to 85% RH (without freezing or condensation)
Protective structure	IP67 (IEC standard) Error prevention circuit

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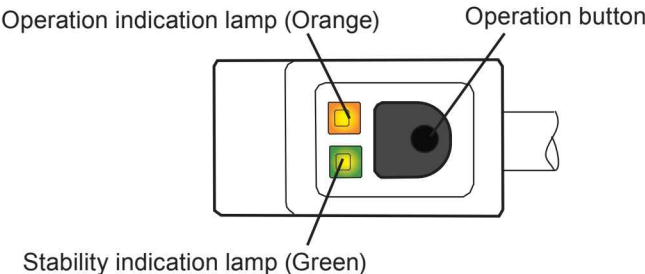
■ Output stage circuit diagram



3.5.2 How to adjust photoelectric sensor unit

Referring to the table below, adjust the photo eye unit (standard type) [TYPE:HP7-P11] after replacing it.

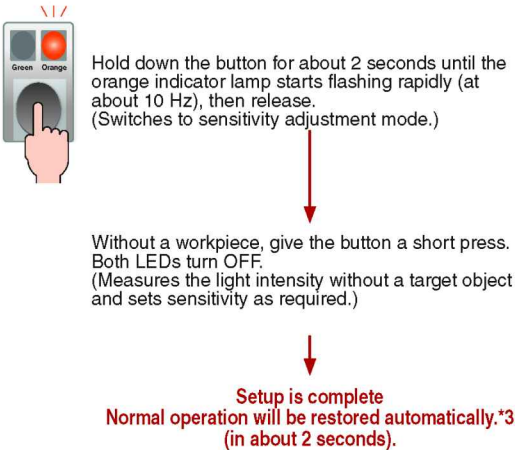
■ Outline view



■ How to adjust sensitivity

■ Tuning without a workpiece

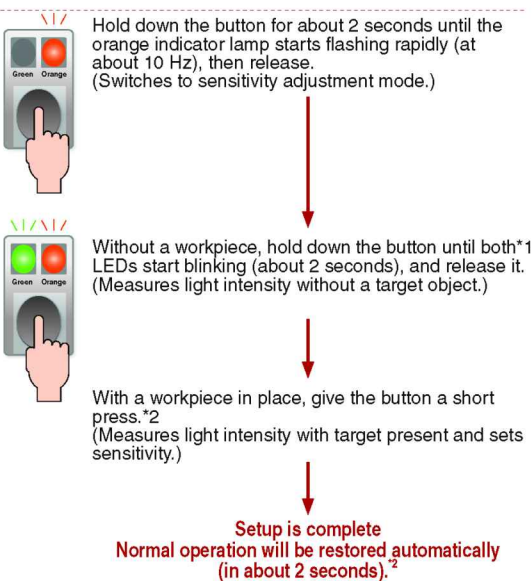
The factory default setting is maximum sensitivity. If the target is not detected consistently at this setting, adjust the sensitivity using the Auto Adjust button as described below.



*3. If the indicator lamp flashes repeatedly, repeat the procedure as described under Indicator lamp flashes repeatedly.

■ 2-point tuning

2-point tuning is used in situations when tuning without a workpiece does not provide a stable detection. Two-point tuning automatically sets the light sensitivity to a value mid-way between the state when the target is present and when the target is absent.

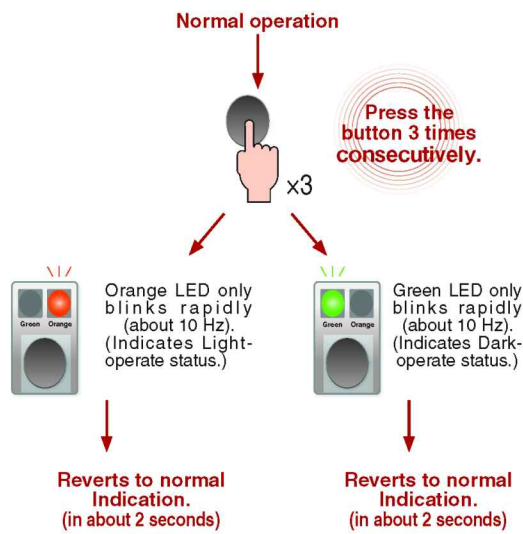


*1. It is OK to reverse the order of the two states (target present/target absent).

*2. If the indicator lamp flashes repeatedly, repeat the procedure as described under Indicator lamp flashes repeatedly

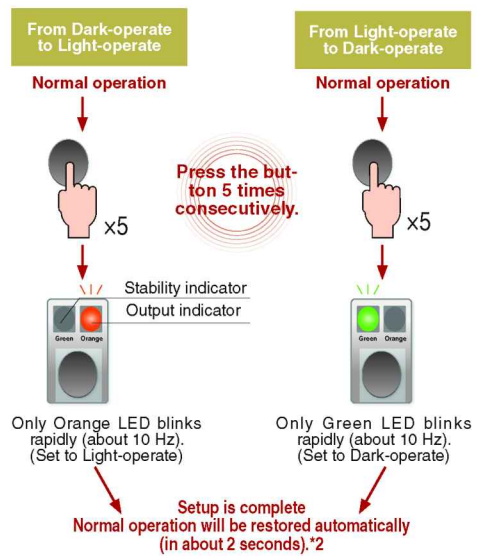
■ Checking LO/DO

Use the procedure shown below to check the current operating mode.



■ LO/DO Changeover

The operating mode is set to default at the factory, but can be changed as outlined below. Light-operate changes to Dark-operate, and Dark-operate changes to Light-operate.



NOTE

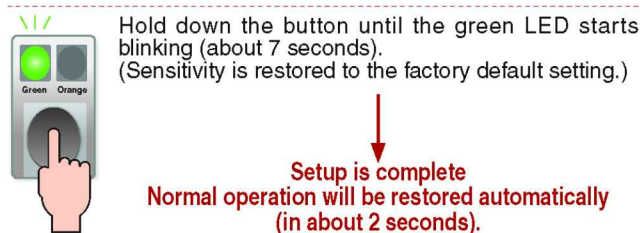
Set the operation selector switch to Light ON when using this equipment.



Different type of the photo eye unit from standard type may be used according to specifications. For the adjustment, refer to the documents provided from the vendor.

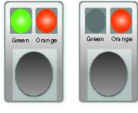
■ When confused, or to restore the default setting (max. sensitivity)

The factory default setting is maximum sensitivity. This procedure is used to restore the factory default setting, which may be useful for resolving confusion during the setting procedure. This procedure will restore sensitivity to the factory default setting from any state (irrespective of how the indicator is flashing).



Indicator lamp flashes repeatedly

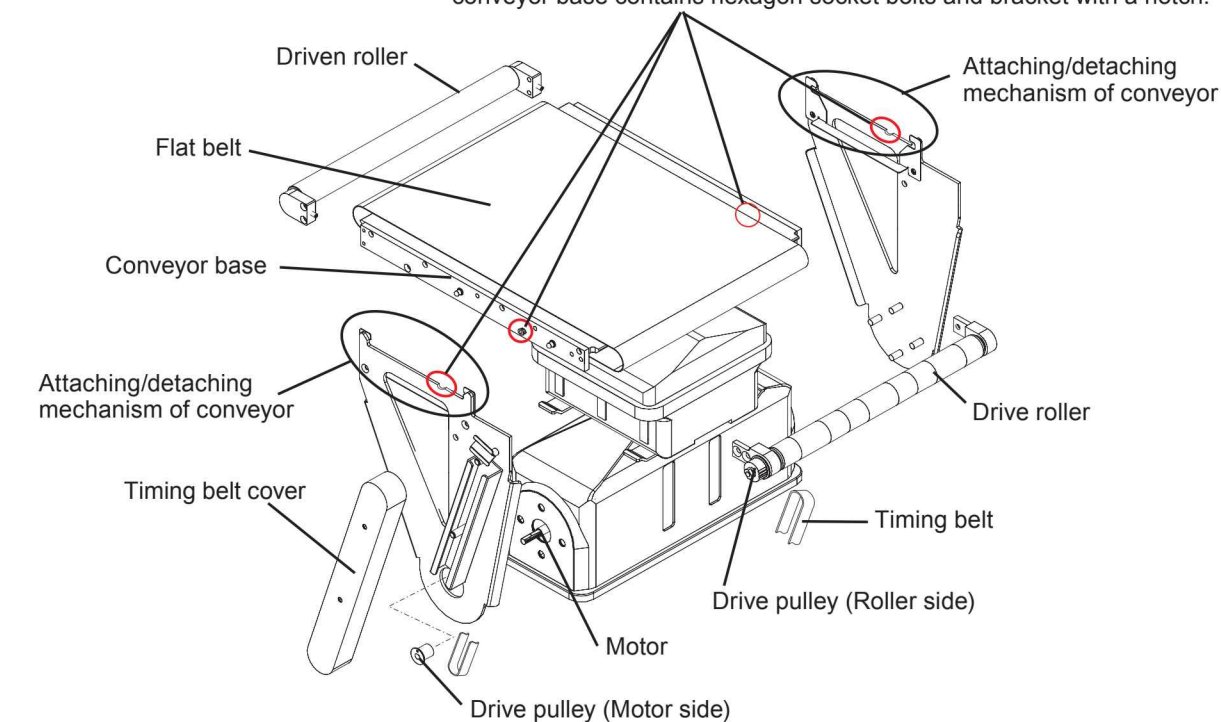
The table below lists the various states indicated by repeated flashing together with suggested responses. If the problem is not resolved, it may be necessary to try a different model of sensor.

LED indicators	Status	Solution
 Orange indicator flashes rapidly or both indicators flash rapidly (at about 10 Hz)	Tuning in progress	Hold down the button until the green indicator flashes rapidly (about 7 seconds) to restore the factory default setting (maximum sensitivity).
 Orange LED only blinks slowly. (at about 1 Hz)	Tuning Without a tuning workpiece Tuning failed - insufficient light	Thru-scan and retroreflective models Press the button once to revert to normal operation at the pre-tuning sensitivity. Adjust the light axis and then repeat the tuning procedure.
	2-point tuning Tuning failed - insufficient light at both points	Press the button once to revert to normal operation at the pre-tuning sensitivity. Thru-scan and retroreflective models Adjust the light axis and then repeat the tuning procedure. Diffuse-scan models Move the sensor closer to the target to boost the reflected light intensity and then repeat the tuning procedure.
	2-point tuning Tuning failed - too much light at both points	Thru-scan models Press the button once to revert to normal operation at the pre-tuning sensitivity. Reduce the amount of light by using slits or tilting the optical axis, and then repeat the tuning procedure.
 Both LEDs blink slowly at the same time. (at about 1 Hz)	Tuning without work-piece Setup is done but light intensity is too high. Stability Indicator may not light up.	Press the button once to revert to normal operation based on the tuning results. Use a workpiece to verify that the sensor works properly. Thru-scan models Reduce the amount of light by mounting slits or tilting the optical axis, and then repeat the tuning procedure. Diffuse-scan models Minimize the reflected light by painting the background black, and then repeat the tuning procedure.
	Tuning without work-piece Setup is done but too low. Stability Indicator may not light up.	Thru-scan and retroreflective models Press the button once to revert to normal operation based on the tuning results. Adjust the light axis and then repeat the tuning procedure.
	2-point tuning Setup is done but Stability indicator may not light up (Not enough difference in light intensity between the two points).	Thru-scan, retroreflective, and diffuse-scan models Press the button once to revert to normal operation based on the tuning results. Check operation before use.

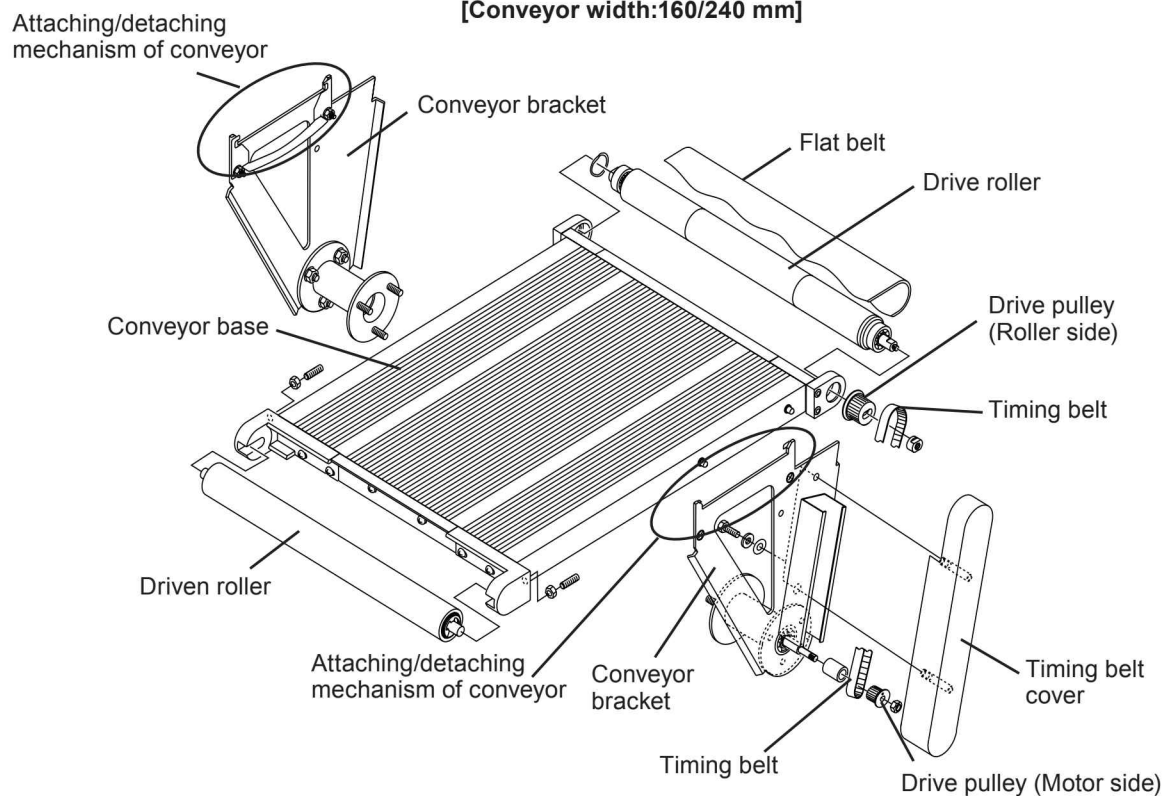
3.6 Weigh Conveyor (SS Type)

■ Outline view of the unit

For machine with the same conveyor size for both infeed and weigher, weight conveyor base contains hexagon socket bolts and bracket with a notch.



[Conveyor width:160/240 mm]



[Conveyor width:320/400 mm]

■ Function of the unit

- Carrying products to weigh conveyor.

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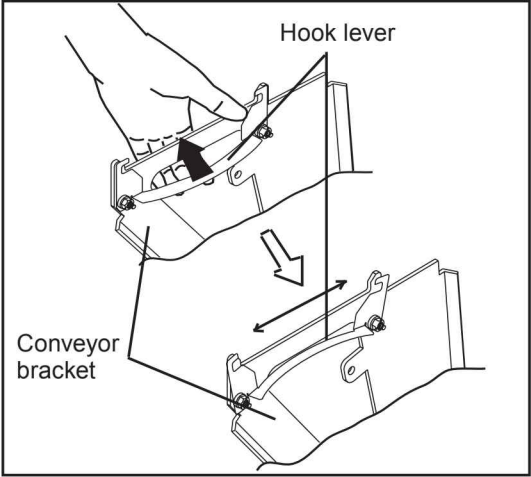
3.6.1 How to detach and attach weigh conveyor

In general, the infeed conveyor is detached and attached at the time of cleaning after operation or during maintenance such as belt replacement. Though it can be easily detached/attached by means of the one-touch detach/attach mechanism, how to detach and attach it is described below for confirmation.

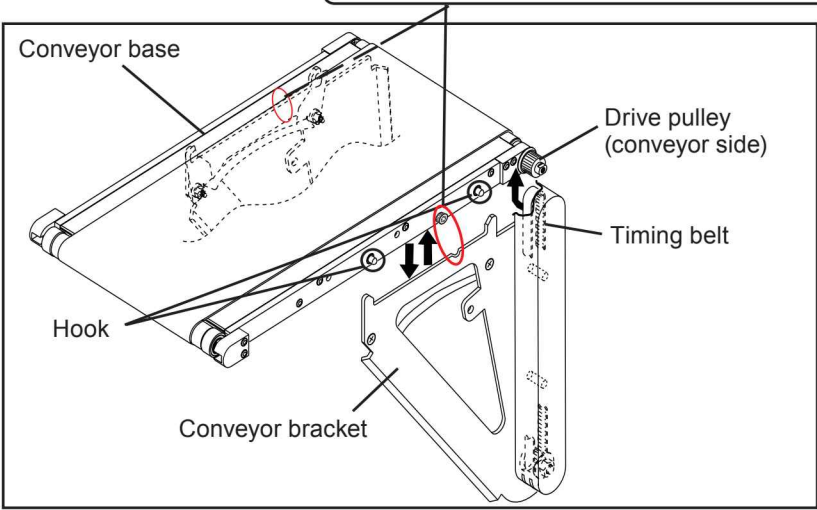
1. Lightly lift up the plate spring of the conveyor bracket with the fingers as shown on the right figure. Then the lever hooks are opened to the right and left. The conveyor base can be now detached. (It can be easily detached by means of the one-touch detach/attach mechanism.)

NOTE

- When detaching the driving pulley (roller side) side, remove the conveyor by pulling out the driving pulley from the timing belt.
- Do not apply excessive load to the weigh conveyor.



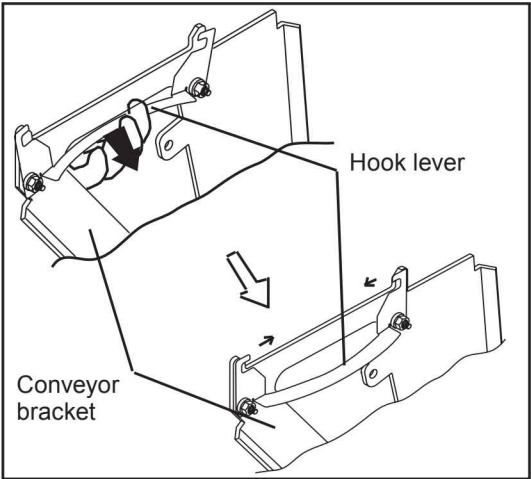
* For machine with the same conveyor size for both infeed and weigher, weight conveyor base contains hexagon socket bolts and bracket with a notch.



2. Place the conveyor base hook on the securing section of the conveyor bracket, and lightly pull down the plate spring with the fingers as shown on the right figure. Then the lever hooks move in the central direction. The conveyor base can be now secured. (It can be easily attached by means of the one touch detach/attach mechanism.)

NOTE

Secure the conveyor base to the conveyor bracket after hanging the driving pulley (roller side) on the timing belt.

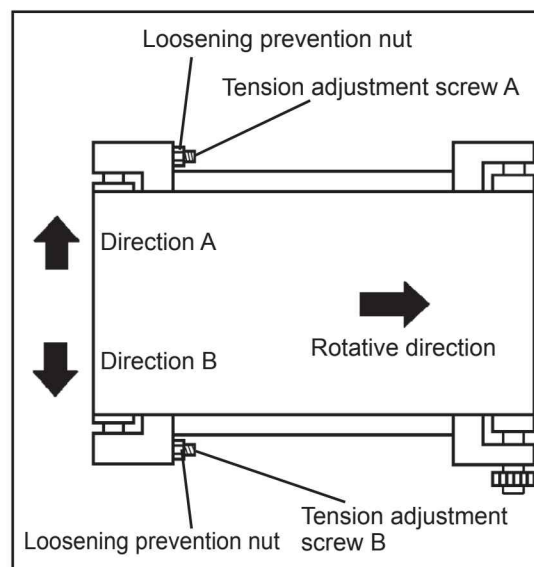


3.6.2 Tension and meandering adjustments of the weigh conveyor belt

The tension of the flat belt for the weigh conveyor is adjusted with the tension adjustment screws. Perform the following steps to adjust the tension of the flat belt. When adjusting the meandering of the flat belt, start from step 3.

1. Make sure that the weigh conveyor is stopped, and tighten tension adjustment screws A and B so that the tension of the flat belt (transfer belt) becomes uniform at both sides to the extent that the driving roller and flat belt (transfer belt) do not run idle.
2. Then tighten the screws by one more turn.

NOTE Make sure that the tension of the flat belt (transfer belt) is uniform at both sides.



3. Set the high belt speed reservation. Touch the operation  key on the main menu of the operation panel.
 - The infeed conveyor starts to operate.
4. Check the inclination direction of the flat belt (transfer belt).
5. When the flat belt (transfer belt) inclines in the A direction, tighten tension adjustment screw A or loosen tension adjustment screw B. When the flat belt (transfer belt) inclines in the B direction tighten tension adjustment screw B or loosen tension adjustment screw A.
6. When inclination of the flat belt (transfer belt) has been removed, lock tension adjustment screws A and B with the loosening prevention nuts. Then, touch the operation  key.
 - The weigh conveyor stops operation.



CAUTION Be careful not to tighten tension adjustment screws A and B too much. The tensile condition in step 2 is optimal and the tension reaches the limit if the screws are tightened by a half more turn.
(If the screws are tightened too much, durability of the bearings extremely becomes worse.)



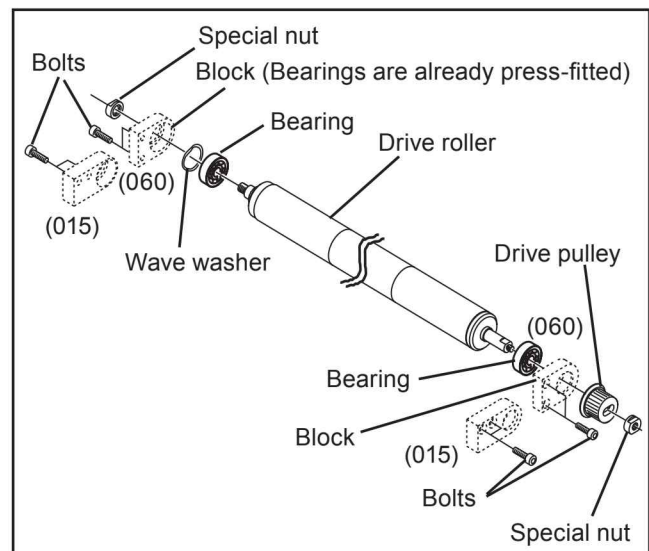
■ Other cautions

- If the tension is not appropriate, the flat belt (transfer belt) motion may become unstable.
- If the tension is not appropriate, the motor may stop due to overload.
- If the tension is not appropriate, the weighing accuracy may become worse.
- If the conveyor encounters a temperature gradient or sudden temperature change (for example, air conditioner's air directly blowing on the flat belt (transfer belt)), the flat belt (transfer belt) motion may become unstable.
- If objects like a broken bag get caught between the conveyor base and the flat belt (transfer belt), the flat belt (transfer belt) motion may become unstable.
- If an offset load is imposed on the flat belt surface of the conveyor (for example, rubbing of the guide), the flat belt (transfer belt) motion may become unstable.

3.6.3 Replacement of driving roller for weigh conveyor

Perform the following steps to replace the driving roller (including bearings) for the weigh conveyor.

1. Remove the screws for securing the blocks at both ends of the driving roller (2 screws for each block).
2. Remove the special nuts at both ends of the driving roller shaft.
3. Remove the driving pulley.
4. Remove the blocks at both ends.



5. Be careful not to lose a waved washer which is contained in one of the blocks. Be sure to insert the waved washer into the block.
6. Attach new bearings and assemble the driving roller of the infeed conveyor in a retrograde order of step 1 to 5.

NOTE

To replace only the bearing, securely attach the new bearing so that the bearing does not move. Fit the block inside to the bearing if necessary. Do not apply any LOCTITE agent on the bearings.

NOTE

- The motion balance of the weigh conveyor driving roller and timing pulley is set up together at the time of assembly. Attach the timing pulley in the original direction.
- When replacing the driving roller, also replace the timing pulley at the same time. If the driving roller and timing pulley of which motion balance is not set up are mistakenly attached, the weight display may not become stable and the accuracy may become worse. Thus, care must be taken when replacing the weigh conveyor driving roller (including timing pulley). (It is not necessary to care about the motion balance when replacing only bearings.)
- Because the motion balance is set up together with the timing pulley, it is necessary to attach the original timing pulley. (It is recommendable to put a mark on the pulley before removing it.)

3.6.4 Replacement of driven roller for weigh conveyor

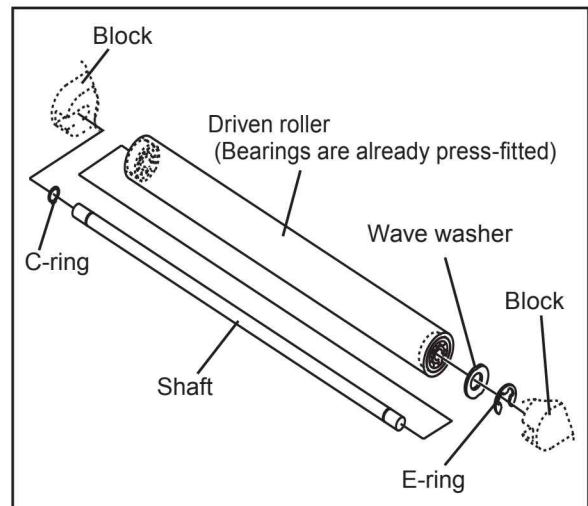
Perform the following steps to replace the driven roller and shaft for the weigh conveyor.

1. Loosen tension adjustment screws A and B and remove the flat belt (transfer belt).

NOTE

Carefully remove the flat belt (transfer belt), because the driven roller sometimes drops out of the block.

2. Remove the C ring of the driven roller shaft, and then the shaft.
3. Attach the shaft removed in step 2 to a new driven roller.
4. Attach the C ring to the driven roller shaft. Attach the driven roller unit to the blocks, and then attach the flat belt (transfer belt). Tighten tension adjustment screws A and B and perform tension and meandering adjustments of the infeed conveyor belt in "3.6.2. Tension and Meandering Adjustments of the Weigh Conveyor Belt".

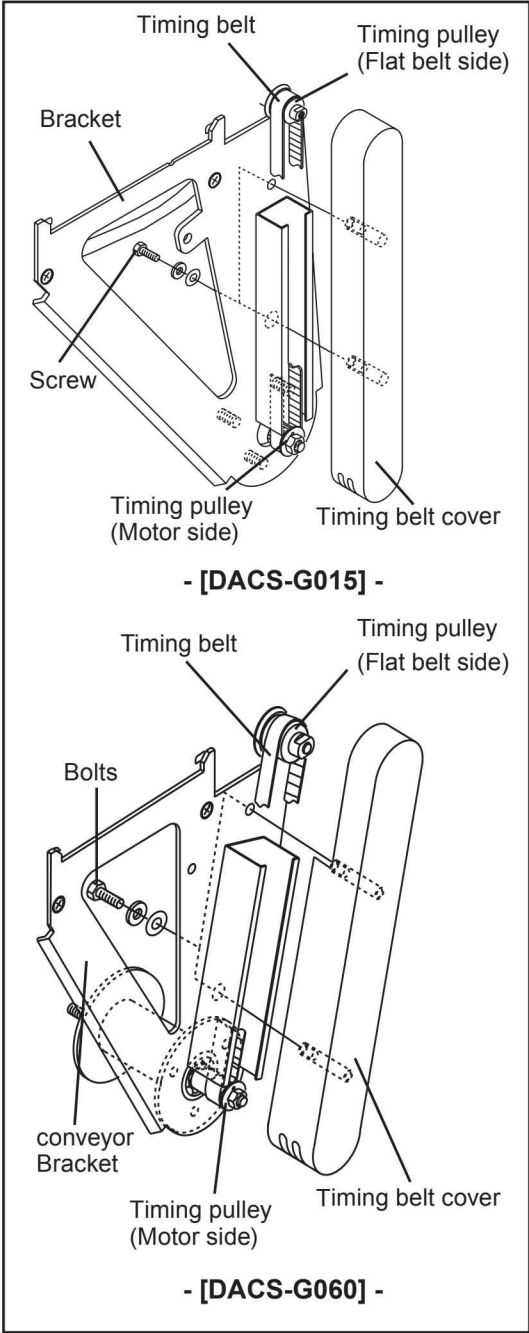
**NOTE**

Because the motion balance of the weigh conveyor driven roller is set up at the time of assembly, care must be taken when replacing it. If the driven roller of which motion balance is not set up is mistakenly attached, the weight display may not become stable and the accuracy may become worse. Thus, care must be taken when replacing the weigh conveyor driven roller.

3.6.5 Replacement of timing belt

In general, the timing belt does not wear so much because it is located inside the stainless cover. However, it may wear out if used over the long term. In such cases, replace the belt according to the following steps.

1. Remove 2 securing screws for the timing belt cover inside the conveyor bracket, and then remove the timing belt cover.
2. Remove the old timing belt.
3. Attach a new timing belt to the driving pulley (flat belt side). At this time, slightly pull down the timing belt and hang it on the driving pulley (motor side).
4. Attach the timing belt cover to the conveyor bracket and secure them with 2 screws

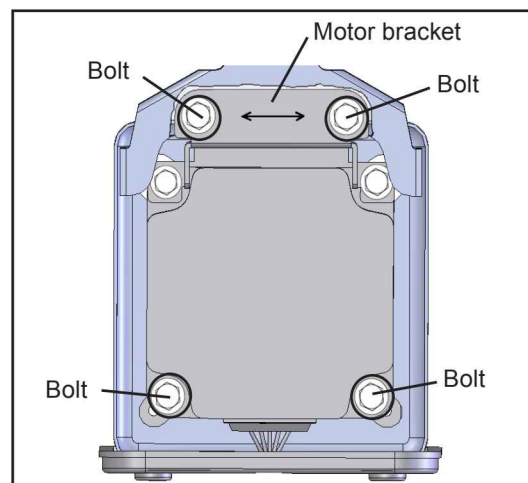


3.6.6 Adjusting tension of timing belt

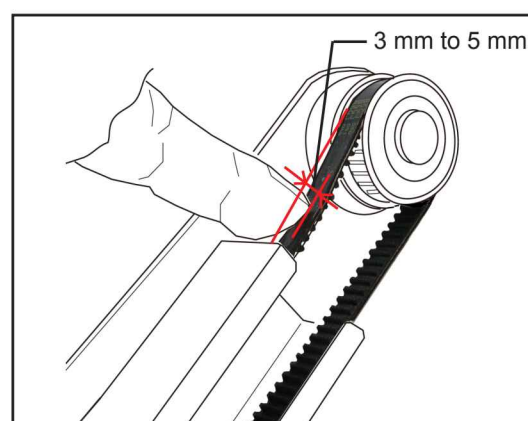
After replacing the timing belt and driving motor, check and adjust the tension of timing belt by the following steps.

■ For the conveyor of 160/240 mm wide (common to SS, CR)

1. Slightly loosen four bolts fixing the motor bracket, move the motor bracket, and adjust the tension of the timing belt.
2. Tighten four bolts with the timing belt tightened.

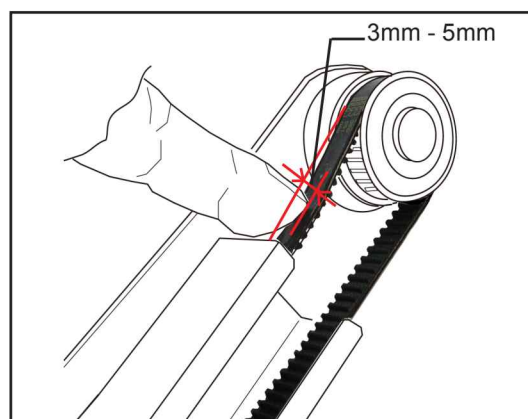
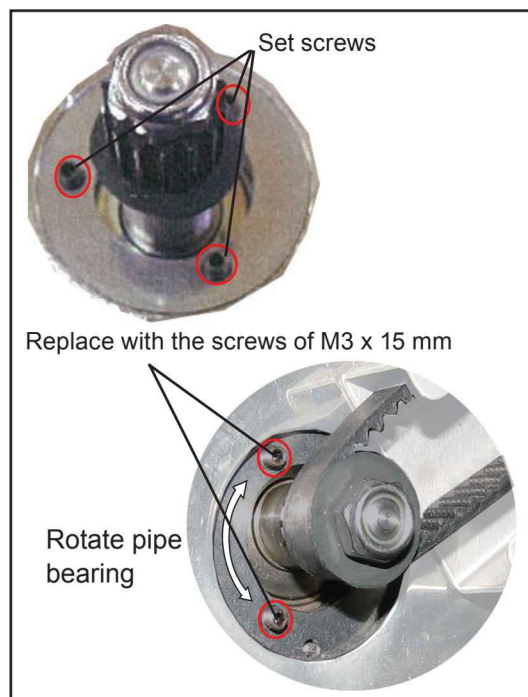


3. Gently press the timing belt at the position shown in the right figure. Check that the timing belt moves approximately 3 mm to 5 mm. If it does not move or moves less/more than the above range, adjust the tension again by referring to the step 1



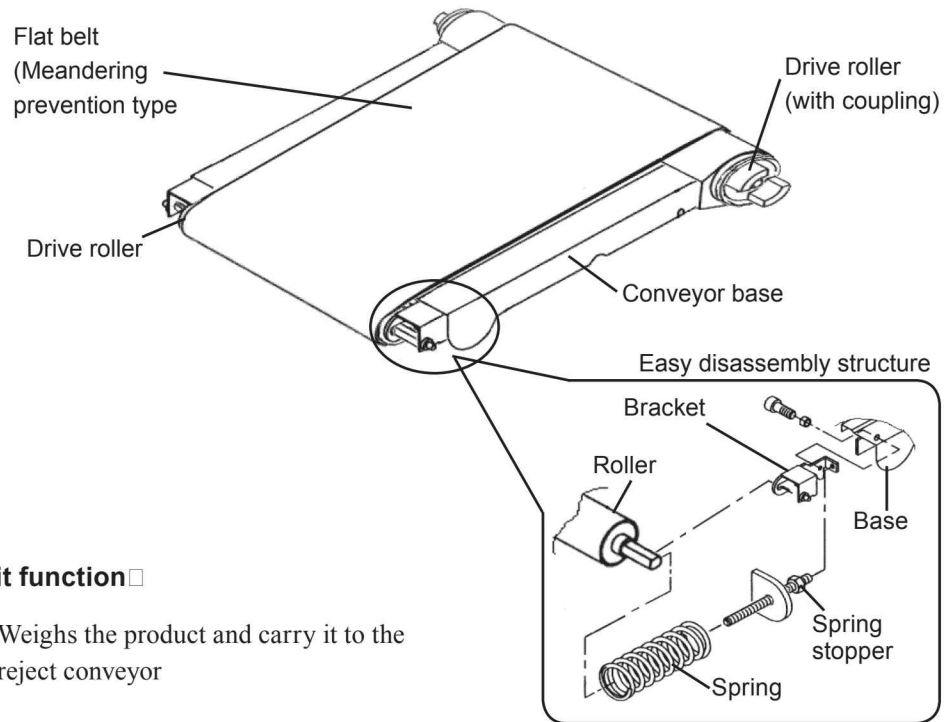
■ For the conveyor of 320/400 mm wide (common to SS, CR)

1. Loosen three set screws fixing the pipe bearing. Remove two set screws, and attach two screws of M3 x 15 mm.
2. Rotate the pipe bearing by using two screws of M3 x 15 mm, to adjust the tension of timing belt.
3. With the timing belt tightened, tighten the set screw. Remove two screws of M3 x 15 mm, attach the set screw, and tighten it.
4. Slightly press the timing belt at the position shown in the right figure. Check that the timing belt moves approximately 3 mm to 5 mm. If it does not move or moves less/more than the above range, adjust the tension again by referring to the steps 1 to 3



3.7 Weigh Conveyor (CR Type)

■ Outline view of the unit



■ Unit function□

- Weighs the product and carry it to the reject conveyor

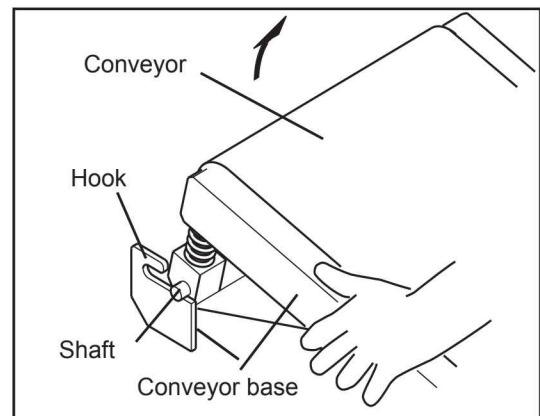


- Since the conveyor belt of CR specification is a meandering prevention type, the meandering adjustment of the standard conveyor belt is not necessary.
- If assembling the conveyor unit of CR specification on site, adjust the length of the conveyor on both sides by means of the one-touch assembly/disassembly mechanism on the driven roller side in order to prevent it from meandering.

3.7.1 Disassembly of Weigh Conveyor

Usually, infeed conveyor is disassembled for cleaning after operation or maintenance such as replacement of belts. Easy disassembly structure enables you to disassemble the conveyor as shown below.

1. Lift up the intake side of weigh conveyor to release the shaft from the hook as shown in the right figure. (Removal can be easily done by the easy disassembly structure.)



NOTE

Do not apply excessive load to the weigh conveyor. The weigh conveyor is directly connected to the weigh sensor.

3 Maintenance Adjustment and Replacement

2. As in the right figure,
 - 1) Lift the discharged side of weigh conveyor to the direction of arrow mark.
 - 2) Lift up the conveyor (drive roller) by pulling it to the direction of arrow mark, and remove the conveyor.

NOTE Be careful not to damage the coupling of the drive roller because it may be a little hard to disassemble it.

3. Remove the weigh conveyor, clean and check it, then assemble it again.
4. Align the coupling notch of the drive roller side of the weigh conveyor with the coupling pin of the drive bearing side.

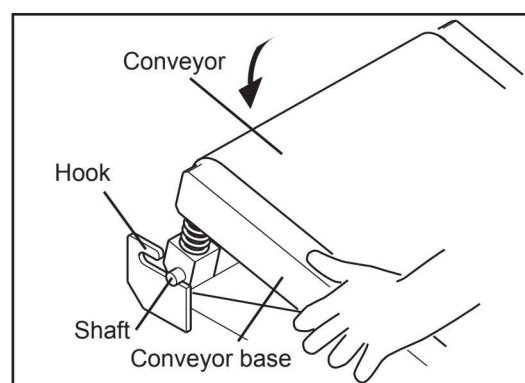
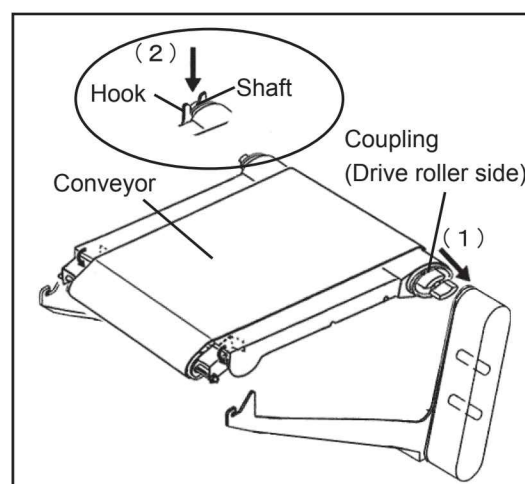
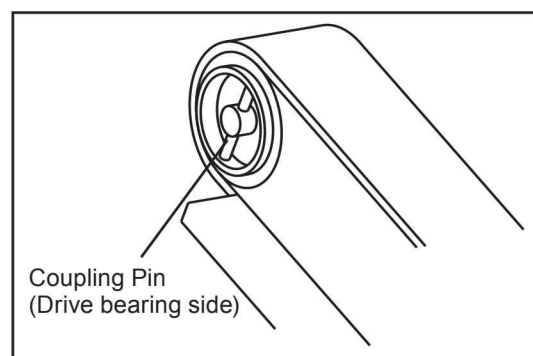
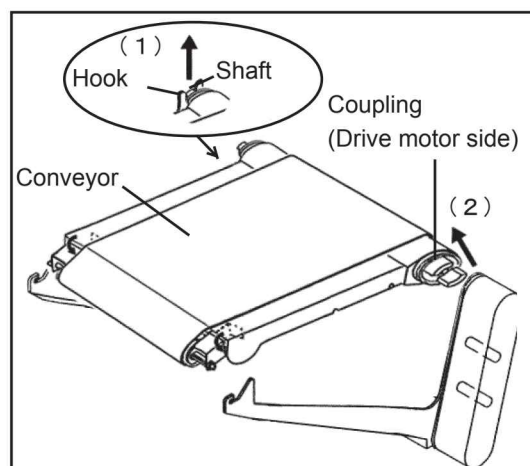
NOTE In assembling the drive roller coupling, ensure correct setting; the coupling pin may be hard to fit the notch of weigh conveyor.

5. Insert the conveyor (drive roller) at the discharged side of the weigh conveyor to the direction (1), then to the direction (2).

NOTE Be sure to hold the conveyor base by your fingers and then push the conveyor base so as not to get your finger caught.

6. Hold the intake side of the weigh conveyor by gripping the conveyor base as shown in the right figure, and press the conveyor. (Easy disassembly structure helps you to attach it easily.)

NOTE Do not apply excessive load to the weigh conveyor. The weigh conveyor is directly connected to the weigh sensor.

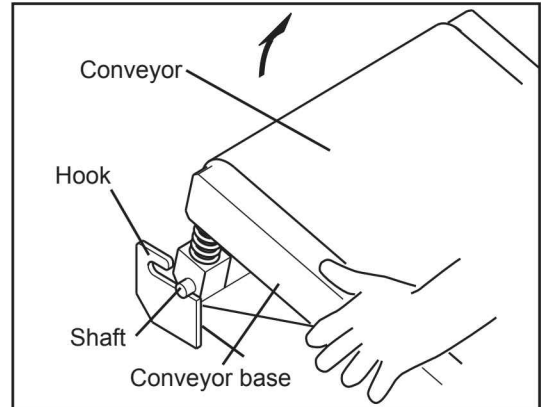


3.7.2 Disassembly (replacement) of Weigh Conveyor Belt

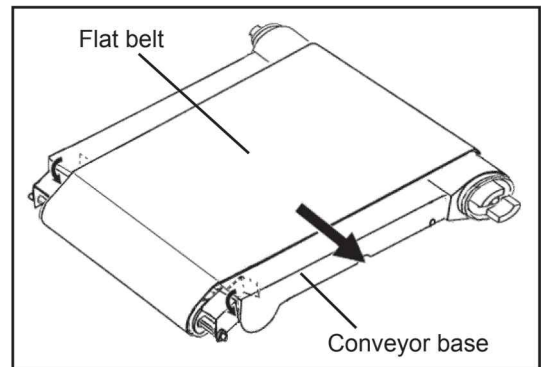
Disassembly (replacement) of weigh Conveyor belt usually, infeed conveyor is disassembled for cleaning after operation or maintenance such as replacement of belts. Easy disassembly structure enables you to disassemble the conveyor as shown below.

1. Remove weigh conveyor by the procedure shown in "3.7.1 Disassembly of Weigh Conveyor".
(It can be disassembled by the easy disassembly structure.)

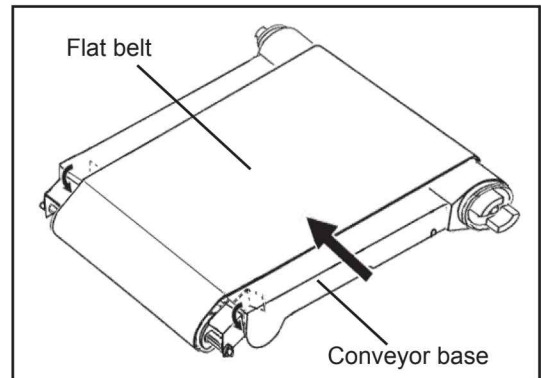
NOTE It may be a little hard to disassemble the coupling of drive roller; take care.



2. From the disassembled weigh conveyor, remove the flat belt to the direction shown in the right figure.



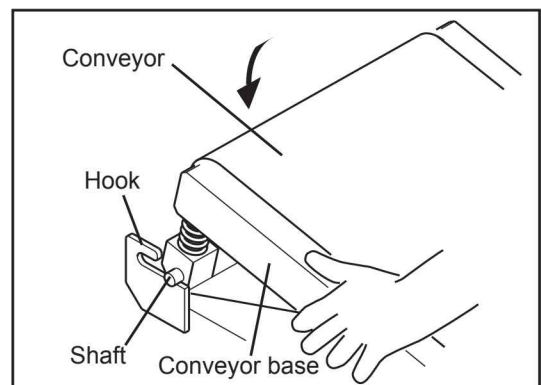
3. Install new flat belt to the weigh conveyor base to the direction as shown in the right figure.



4. Attach the weigh conveyor by the same procedure shown in "3.7.1 Disassembly of Weigh Conveyor".
(Easy disassembly structure helps you to attach it easily.)

NOTE

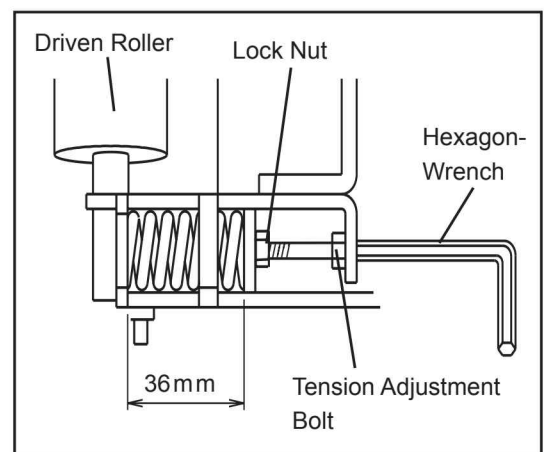
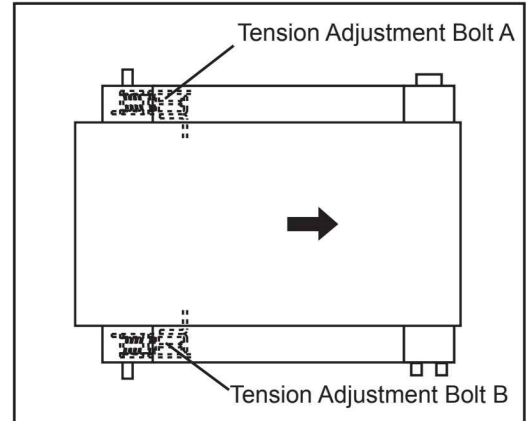
- When pushing the shaft at the conveyor end in the hook, hold the conveyor base with a finger push it in carefully taking care not to be pinched.
- Do not apply excessive load to the weigh conveyor. The weigh conveyor is directly connected to the weigh sensor.



3.7.3 Tension and meandering (snaking) adjustment of the weigh conveyor

The tension of the flat belt for weigh conveyor must be adjusted by the adjustment bolt. Follow the instruction below for the tension adjustment of the flat belt. In case of adjustment of the lateral shifting for the flat belt, follow the instruction from 2.

1. Adjust the spring length to 36 mm by adjustment bolt at the driven roller side of infeed conveyor.
2. Press  key to make infeed conveyor active. Check and see if flat belt is in the proper (center) position. Then, press  key to complete the tension adjustment.
3. Turn the tension adjusting bolts A/B equally to adjust the tension of the conveyor belt.
4. Make sure that the conveyor guide fit the roller's groove and no excessive force is applied.



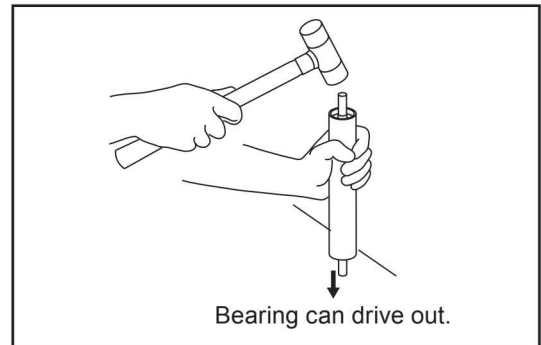
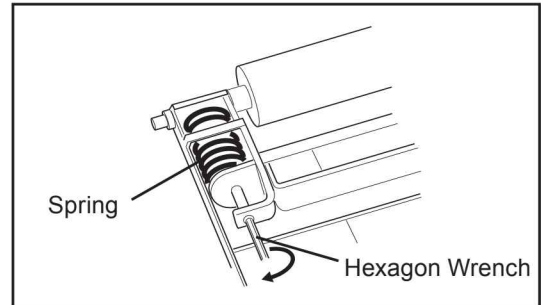
NOTE

- Since the conveyor belt of CR specification is a meandering prevention type, the meandering adjustment of the standard conveyor belt is not necessary.
- Still, if the right/left tension adjustment is poor, the meandering prevention guide of flat belt becomes worn to cause powder swarf.
- Adjust the length of the conveyor on both sides by means of the one-touch assembly/disassembly mechanism on the driven roller side in order to prevent it from meandering.

3.7.4 Replacement of driven roller bearing (and oil seal) for conveyor

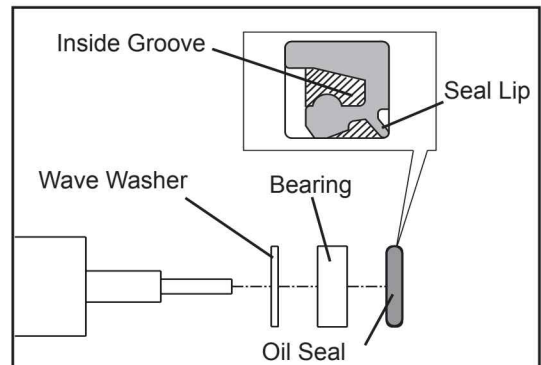
For Replacement of driven roller bearing (and oil seal) for conveyor, follow the procedure

1. Unfasten (rotate clockwise) the set screw by hexagon wrench to decrease the spring force and remove the spring that fixes the both ends of driven roller.
2. Remove the driven roller, and remove the oil seal from the roller shaft.
3. Tap the roller shaft with a plastic hammer to drive out the bearing at one end.



NOTE Do not excessively hit the bearing. Otherwise, it will be damaged.

4. Drive out the other bearing from the roller using a round bar.
5. With the reverse procedure to above, assemble the wave washer, the bearing and the oil seal, and attach the driven roller to the conveyor.



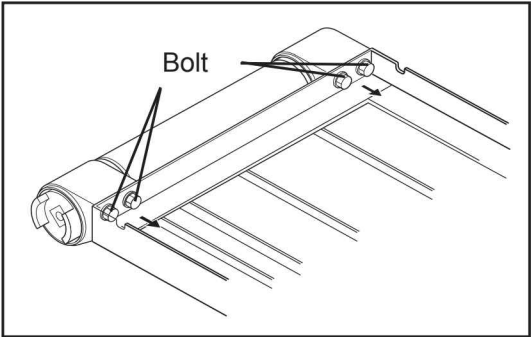
NOTE

- When mounting the oil seal, sufficiently apply the specified grease to the inside groove and the seal lip. (The shaded area must be greased.)
- Specified grease: Kluber Lubrication, PARALIQ (R) GTE703

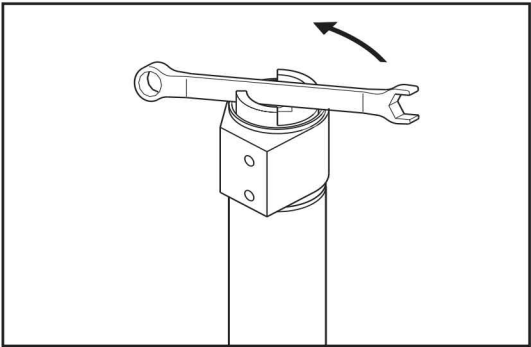
3.7.5 Replacement of belt-side bearing (and oil seal) for conveyor driven roller

For replacement of belt-side bearing (and oil seal) for conveyor drive roller, follow the procedures below:

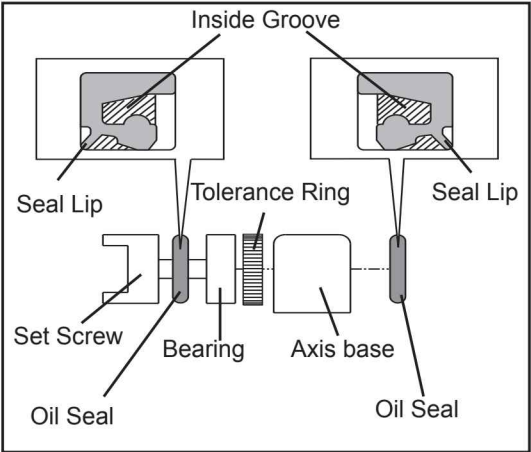
1. Take out 4 bearing fixing bolts and remove the bearing of drive roller.
2. Rotate the set screw of the bearing at coupling side counterclockwise as shown at right, and remove the bearing.



3. Insert a screwdriver into the shaft hole to take 2 oil seals out of the bearing.
4. Strike the bearing by a round bar to remove the bearing.



5. By the reverse procedure to above, assemble a bearing and an oil seal, and attach the bearing to the conveyor.



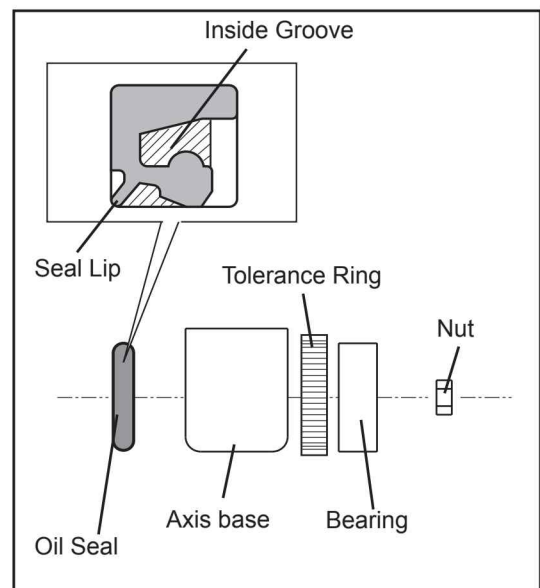
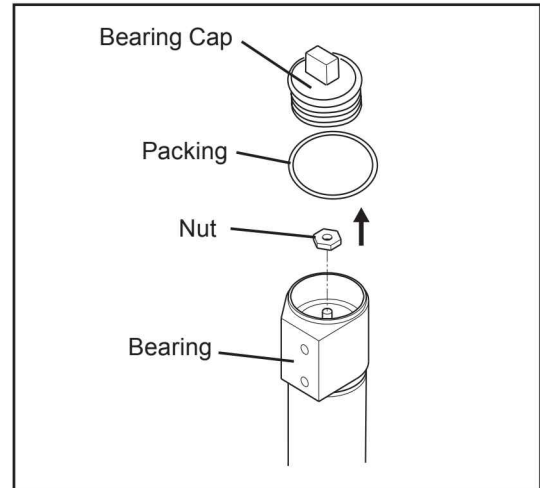
NOTE

- When mounting the oil seal, sufficiently apply the specified grease to the inside groove and the seal lip. (The shaded area must be greased.)
- Specified grease: Kluber Lubrication, PARALIQ (R) GTE703

3.7.6 Replacing Bearing (Oil Seal) of the Conveyor Drive Roller at Anti-coupling Side

Follow the instruction below for replacing the bearing (oil seal) of the conveyor drive roller at anticoupling side.

1. Take out 4 bearing fixing bolts and remove the bearing of drive roller.
2. Remove the bearing fixing the driven roller.
3. Remove the nut inside the bearing to remove the bearing.
4. Insert a screwdriver in the shaft hole and take 2 oil seals out of the bearing.
5. Strike out the bearing by a round bar to remove the bearing.
6. With reverse procedure to the above, assemble a bearing and an oil seal, and attach the bearing to the conveyor.

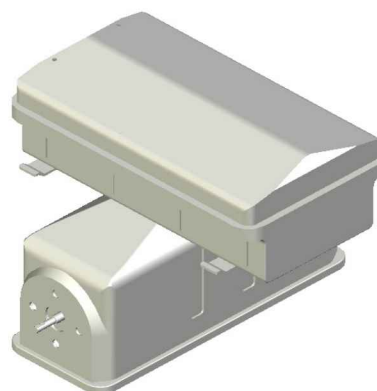


NOTE

- When mounting the oil seal, sufficiently apply the specified grease to the inside groove and the seal lip. (The shaded area must be greased.)
- Specified grease: Kluber Lubrication, PARALIQ (R) GTE703

3.8 Weigh/Drive Unit (SS Type)

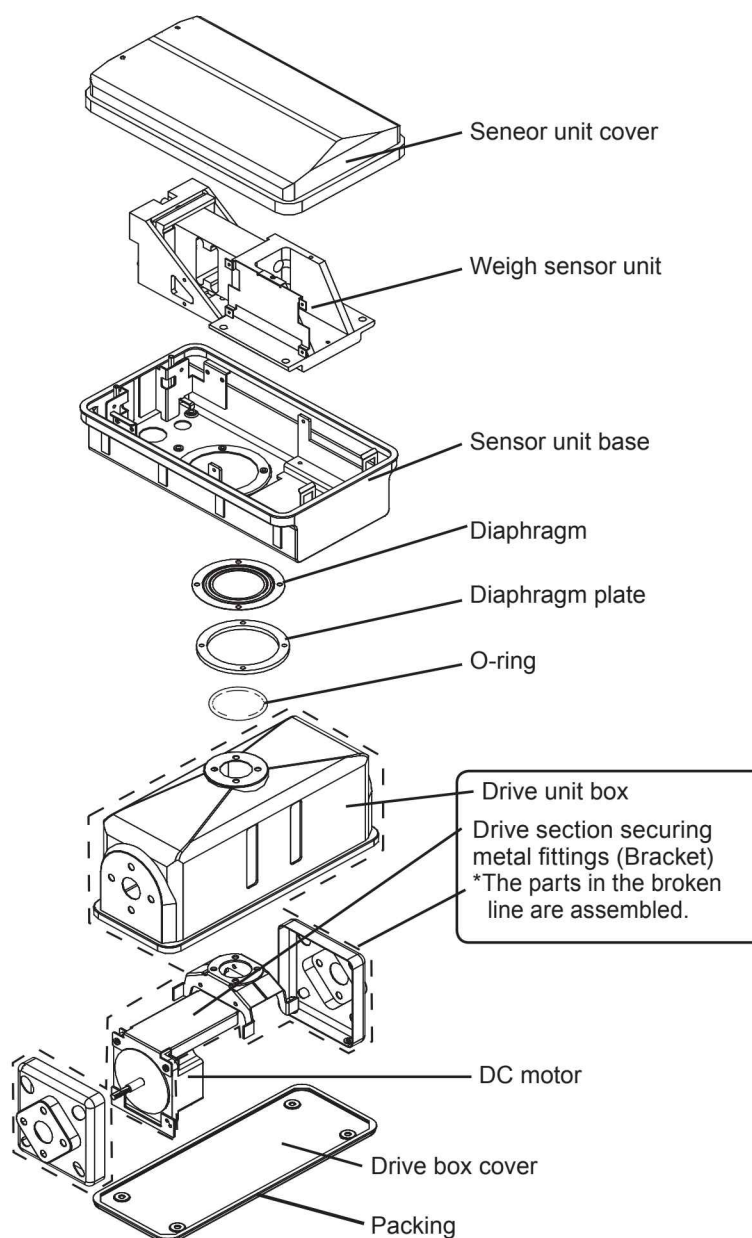
■ Outline view of the unit



■ Function of the unit

- Transfers the motor drive to the weigh conveyor drive roller.
- Weighs the product.

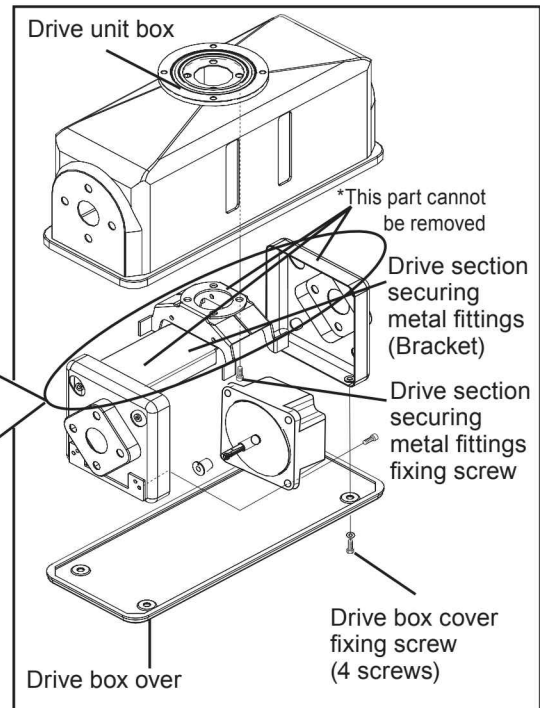
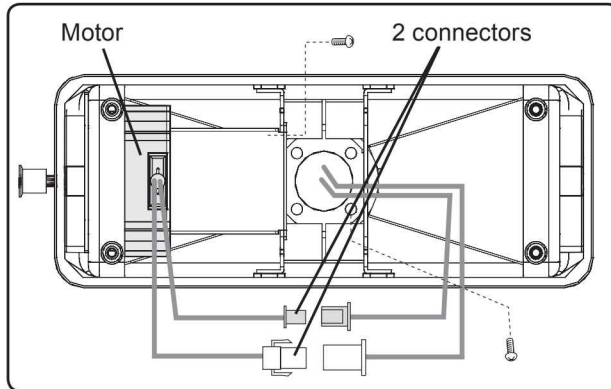
■ Exploded view of the unit



3.8.1 Replacement of drive motor

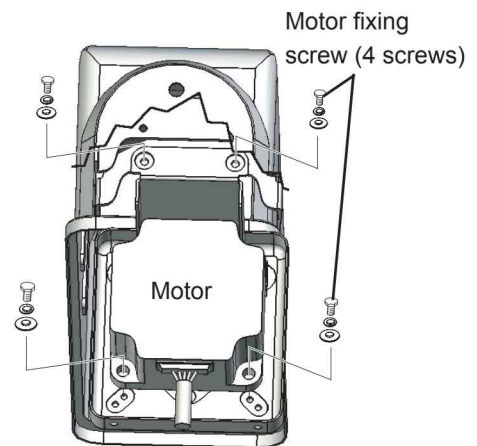
Perform the following steps to replace the drive motor of the infeed conveyor, drive shaft (015 type) bearings.

1. Remove the 4 drive box cover fixing screws at the bottom and remove the drive box cover.
2. Remove the 2 motor cable connectors.

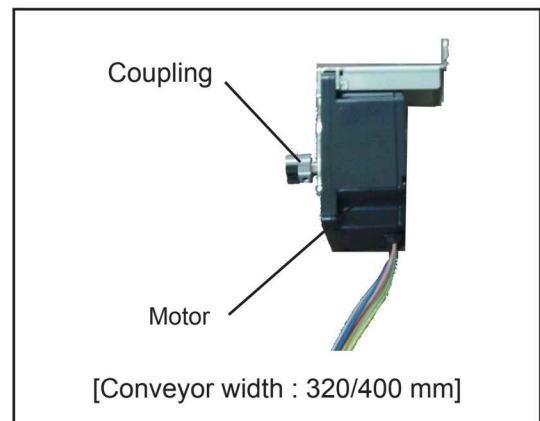


NOTE When removing 4 screws, do not apply excessive load to the weigh conveyor.

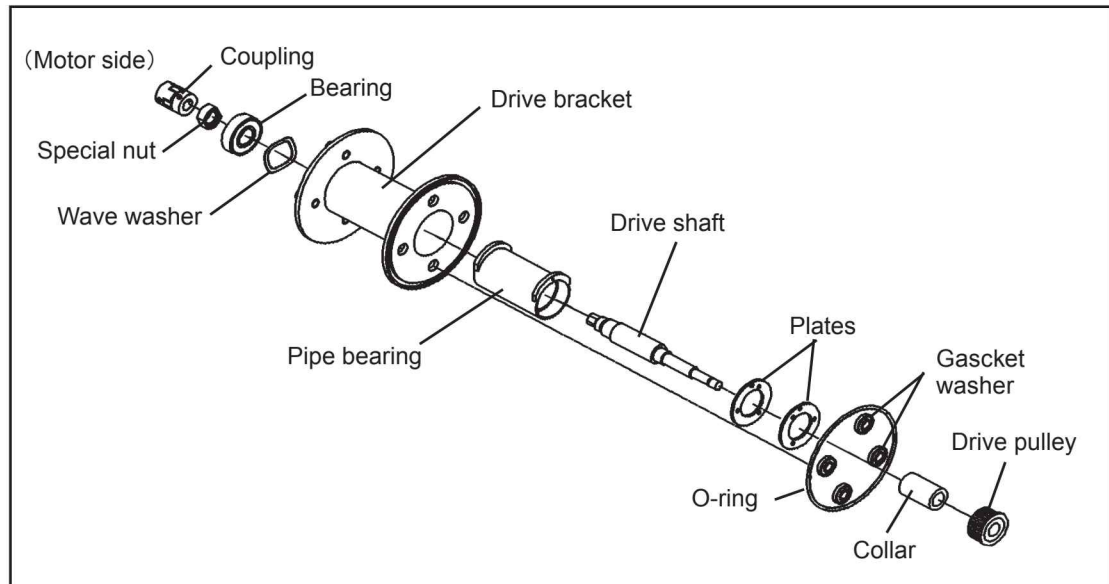
3. Remove 4 drive box securing screws and then remove the motor unit.
4. Remove the timing belt cover of the conveyor bracket, and then loosen and remove the lock nut located on the drive shaft (motor side). Then remove the driving pulley (motor side).
5. Remove 4 motor securing screws and then remove the motor.
6. Attach a new motor and assemble the components in a retrograde order of step 1 to 4.



NOTE When replacing the motor of the conveyor width 320/400 mm machine, note that a coupling is contained in the motor drive shaft.



7. After removing the motor, remove 4 nuts that secure the drive shaft unit inside the drive box, and then remove the drive shaft unit keeping the conveyor bracket attached.
8. Remove 3 nuts that secure the conveyor bracket, and then the conveyor bracket.



9. Loosen the special nut for the drive shaft unit as shown on the figure, and then remove the nut.

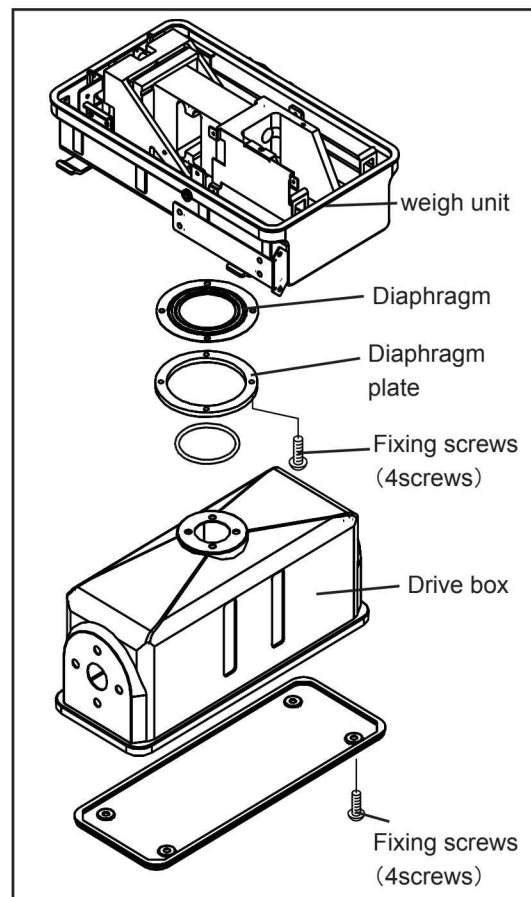
NOTE Be sure to set the special nut in the proper direction.

10. Remove the bearing (motor side).
11. Tap out the drive shaft to the pulley side with a plastic hammer and remove the drive shaft.
12. Remove the bearing (pulley side).
13. Set a new bearing (pulley side). At this time, insert a tolerance ring into the drive bracket and then press-fit a bearing.
14. Assemble the components in a retrograde order of step 7 to 12.

3.8.2 Replacement of diaphragm

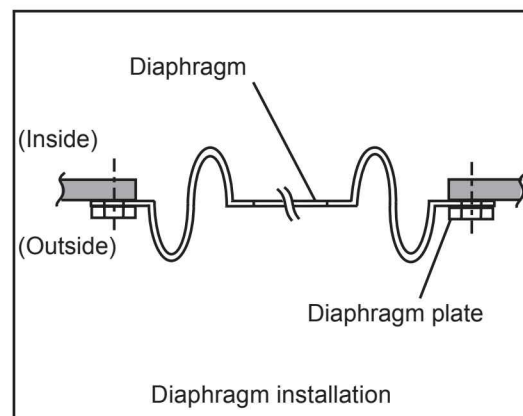
Perform the following steps to replace the diaphragm for the drive section of the weigh conveyor.

1. Remove the drive motor box in a similar manner to the replacement of the drive motor in 3.8.1.
2. As shown on the right figure, remove 4 securing nuts that secure the drive section securing metal fittings, and then remove the fittings.
3. Remove 4 securing screws that secure the diaphragm retainer, and then remove the retainer.
4. Attach a new diaphragm.
5. Assemble the components in a retrograde order of step 1 to 2.



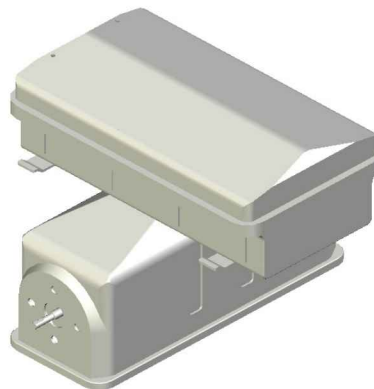
NOTE

- Attach the diaphragm as shown on the right figure. Take care not to make reverse mounting. Doing so will adversely affect the precision.
- If the diaphragm securing screws A are too long, it may hit the weigh sensor to damage, take care.



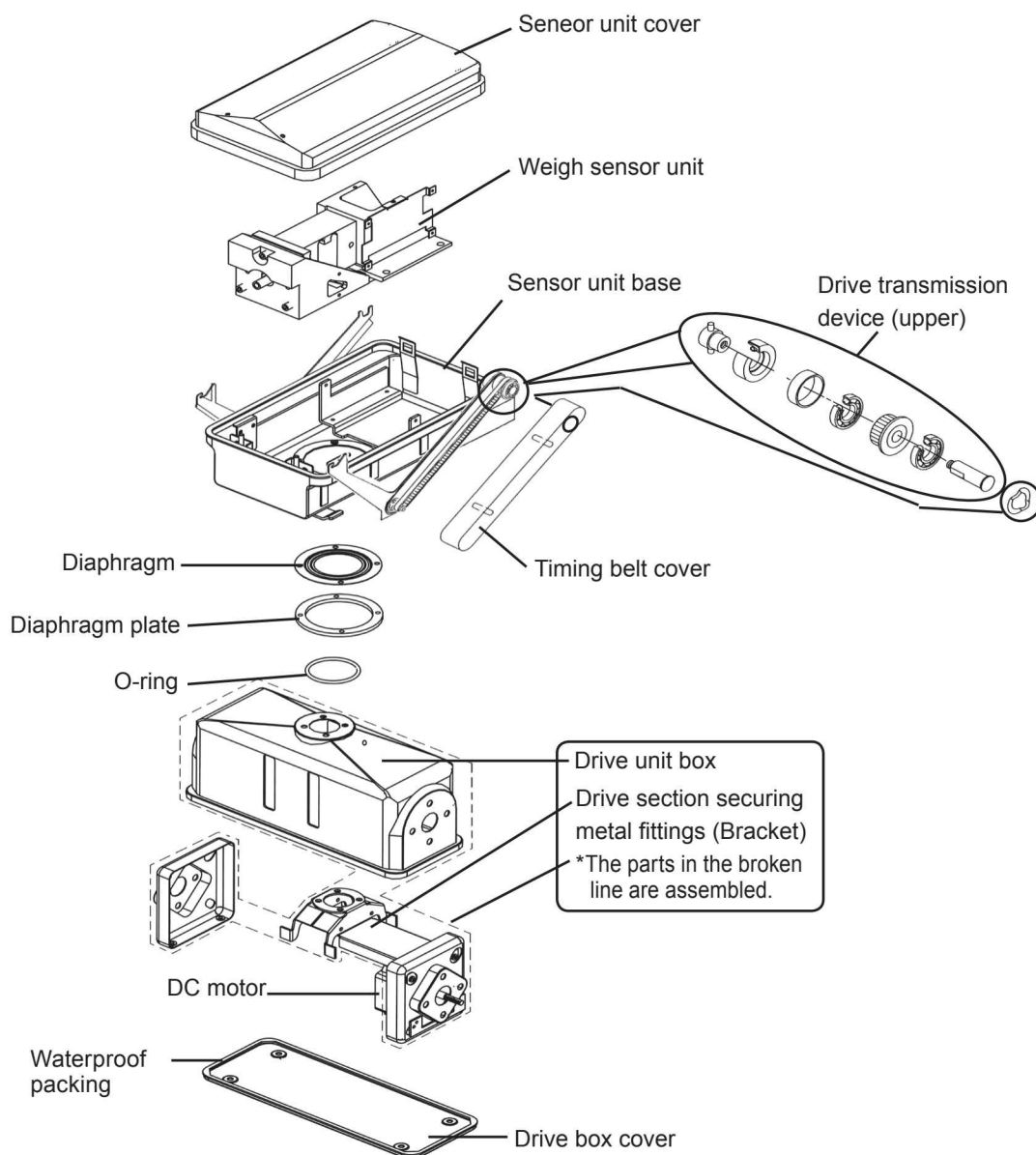
3.9 Weigh/Drive Unit (CR Type)

■ Outline view of the unit



■ Function of the unit

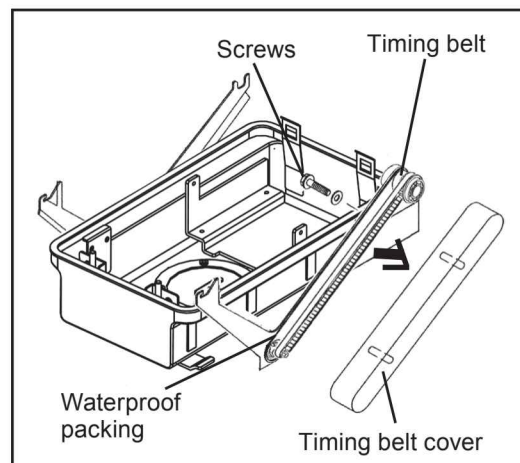
- Transfers the motor drive to the weigh conveyor drive roller.
- Weighs the product.



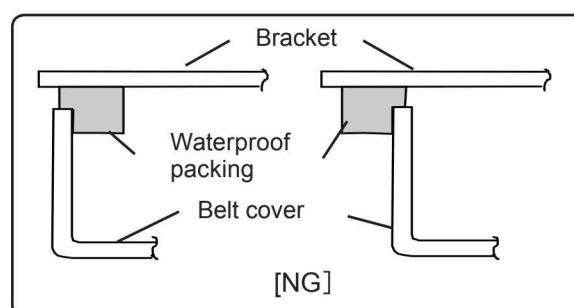
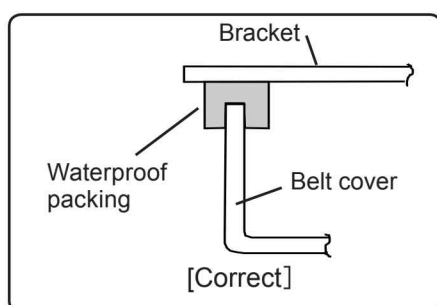
3.9.1 Replacing Timing Belt

Usually, timing belt is not consumed much because it is in the stainless steel cover. However, it may be deteriorated according to the long time use. Following is the procedure of replacement.

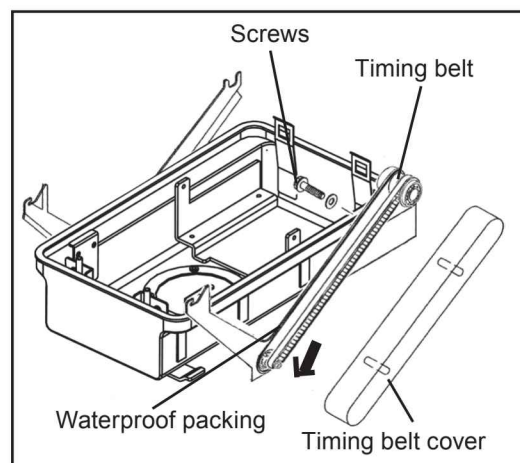
1. Remove the weigh conveyor. Then, remove 2 screws (some models have 3 screws) that fix the timing belt cover.
2. Remove the old belt. Replace the old waterproof packing with a new one.



- After a long time use, the waterproof packing is possibly deteriorated severely. Continual use of the deteriorated packing damages the waterproof effect. When opening the rear plate, replace also the waterproof packing.
- If the belt cover is incorrectly attached as in the example below, expected waterproof effect is not assured; ensure to attach the cover correctly.



3. Set a new timing belt around the upper timing pulley as shown at right, then pull the timing belt to the direction of the arrow at right and catch it on the lower timing pulley.
4. Attach the timing belt cover and fix it with the 2 screws (some models have 3 screws).



3.9.2 Inspection of Upper and Lower Drive Transmission Devices

Usually, the upper and the lower drive transmission devices work silently and smoothly. However, with breakdown of some parts such as bearing, metal sound may come out and the devices do not work smoothly. In that case, the upper and the lower drive transmission devices must be checked.

1. For the upper drive transmission devices, remove the shaft as shown at right and check the bearing. Replace the bearing if necessary.



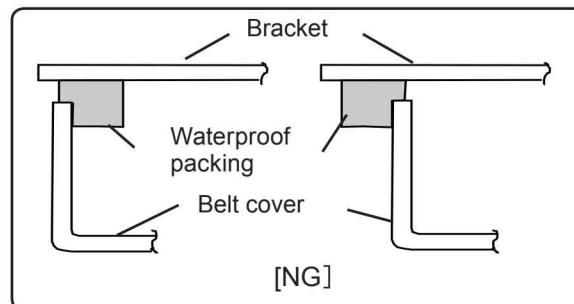
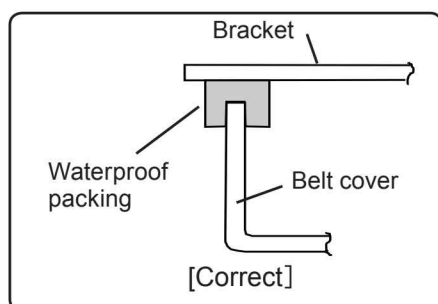
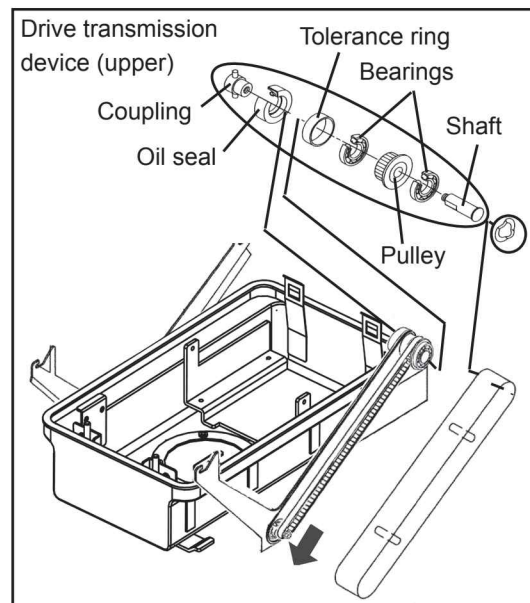
REFERENCE

See 3.9.3 in the next page for the replacement procedure.)



CAUTION

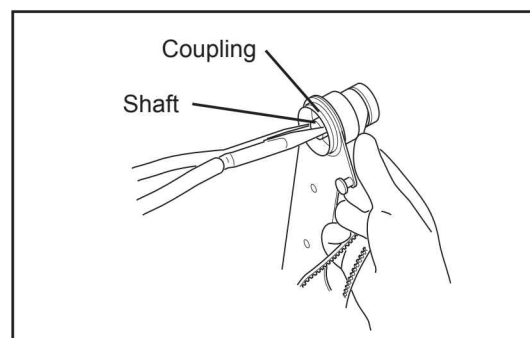
- After a long time use, the waterproof packing is possibly deteriorated severely. Continual use of the deteriorated packing damages the waterproof effect. When opening the rear plate, replace also the waterproof packing. (The waterproof packing is typically to be replaced at opening the cover after an year or longer or when several times of open/close movement of the cover.)
- If the belt cover is incorrectly attached as in the example below, expected waterproof effect is not assured; ensure to attach the cover correctly.



3.9.3 Replacing Bearing (Oil Seal) for Coupling of Upper Drive Transmission Device

Follow the instruction below to replace the bearing (oil seal) for the coupling of upper drive transmission device.

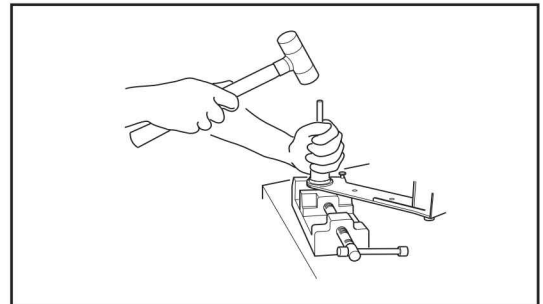
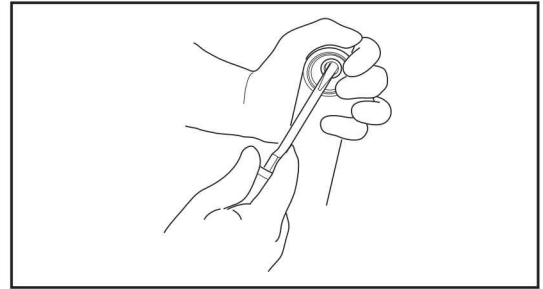
1. Remove 4 screws that are fixed on the load cell bracket, and remove the drive transmission device bracket.
2. Loosen the coupling by using pliers while holding the coupling with the timing belt.
3. Remove the shaft, the gear and the bearing in order.



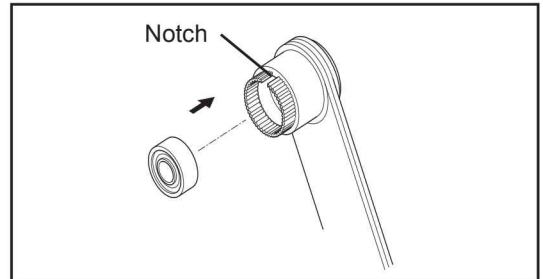
NOTE

Before removing the drive shaft, always lock the shaft with the timing belt so that it will not turn. If the coupling is removed without locking the shaft rotation, pulley may be broken.

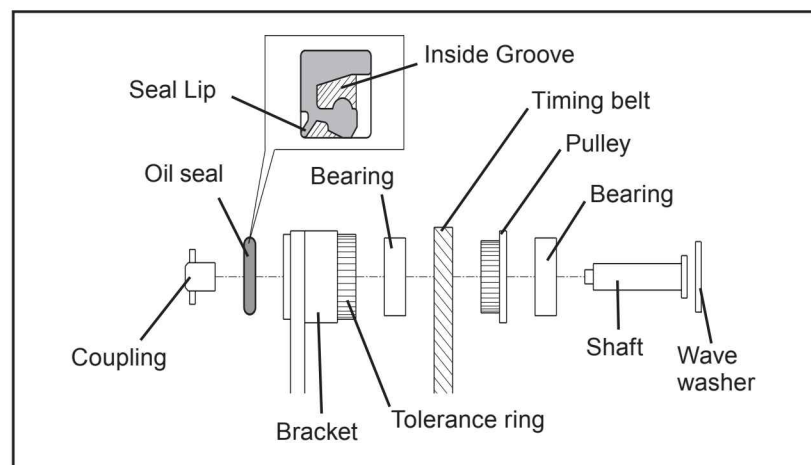
4. Remove the oil seal from inside the bracket.
5. Fix the bracket as shown in the figure and drive out the bearing with a plastic hammer and a round bar.

**CAUTION**

When mounting the tolerance ring, orient the cutout outward, or opposite to the direction that the tractive force of timing belt is applied. When mounting the oil seal, push the oil seal evenly along the periphery to avoid distortion.



6. With the reverse procedure to the above, assemble the bearing and oil seal, and attach the bearing to the conveyor.

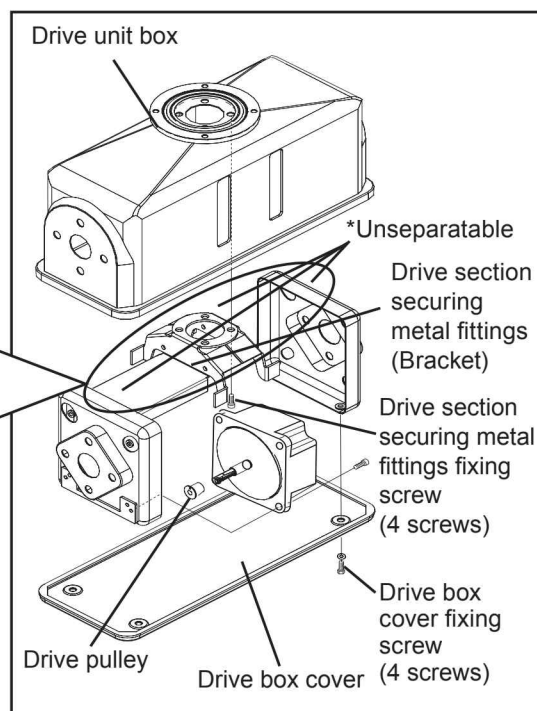
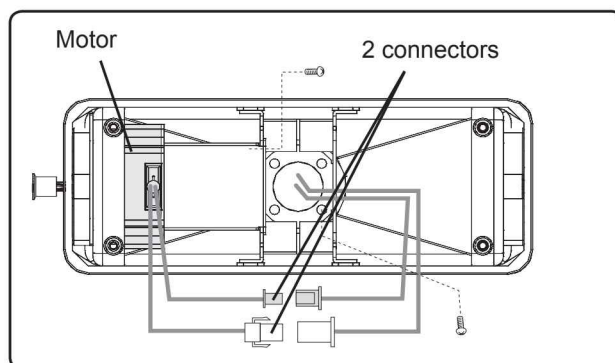
**NOTE**

- When mounting the oil seal, sufficiently apply the specified grease to the inside groove and the seal lip. (The shaded area must be greased.)
- Specified grease: Kluber Lubrication PARALIQ(R) GTE703

3.9.4 Replacement of drive motor

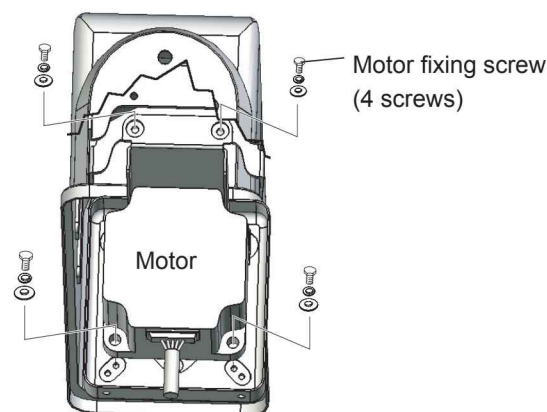
Perform the following steps to replace the drive motor of the infeed conveyor, drive shaft (015 type) bearings.

1. Remove the 4 drive box cover fixing screws at the bottom and remove the drive box cover.
2. Remove the 2 motor cable connectors



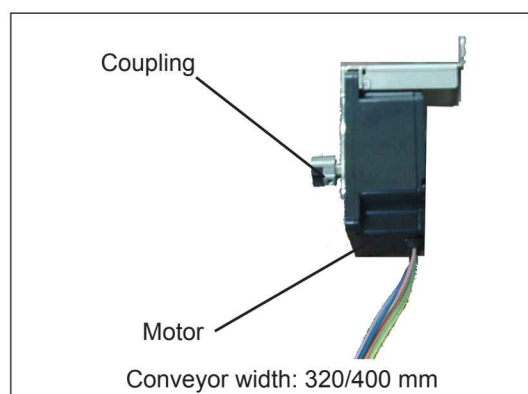
NOTE When removing 4 screws, do not apply excessive load to the weigh conveyor.

3. Remove 4 drive box securing screws and then remove the motor unit.
4. Remove the timing belt cover of the conveyor bracket, and then loosen and remove the lock nut located on the drive shaft (motor side). Then remove the driving pulley (motor side).
5. Remove 4 motor securing screws and then remove the motor.
6. Attach a new motor and assemble the components in a retrograde order of step 1 to 4.

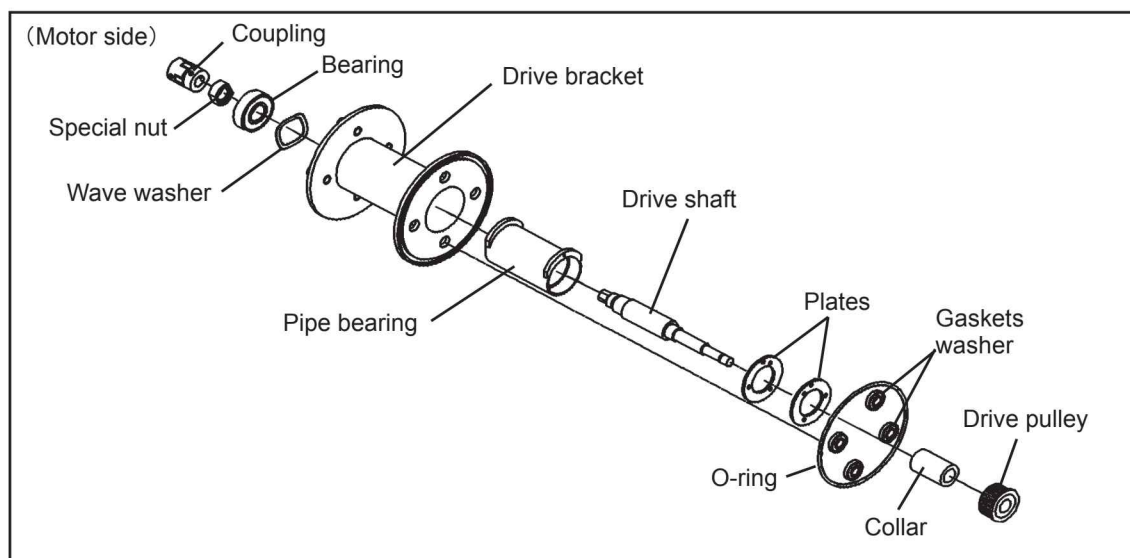


NOTE

- When replacing the motor of the conveyor width 320/400 mm machine, note that a coupling is contained in the motor drive shaft.
- Perform wiring with care not to allow the motor cable contact to the other metal fittings, otherwise it could affect the weighing accuracy.



7. After removing the motor, remove 4 nuts that secure the drive shaft unit inside the drive box, and then remove the drive shaft unit keeping the conveyor bracket attached.
8. Remove 3 nuts that secure the conveyor bracket, and then the conveyor bracket



9. Loosen the special nut for the drive shaft unit as shown on the figure, and then remove the nut.

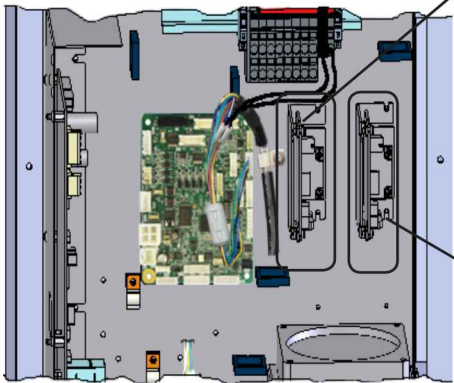
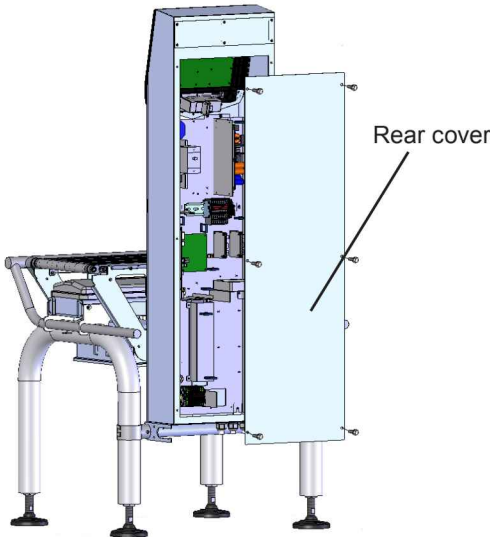
NOTE Be sure to set the special nut in the proper direction.

10. Remove the bearing (motor side).
11. Tap out the drive shaft to the pulley side with a plastic hammer and remove the drive shaft.
12. Remove the bearing (pulley side).
13. Set a new bearing (pulley side). At this time, insert a tolerance ring into the drive bracket and then press-fit a bearing.
14. Assemble the components in a retrograde order of step 7 to 12.

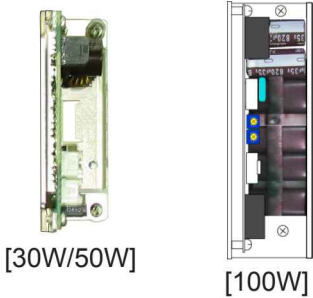
3.9.5 How to Drive the Conveyor Motor in an Emergency

If DACS-GN main unit cannot operate the weight check owing to problems in the weigh unit, follow the procedures below when driving the infeed/weigh conveyors forcibly.

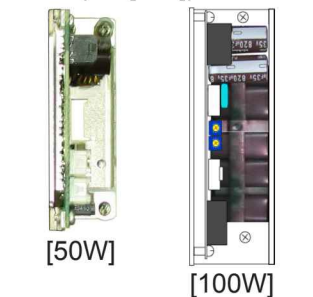
1. Turn the power switch OFF
2. Remove the rear cover attached to the electrical control unit
3. Remove the cables (WTC-CN2,IFC-CN2) on the DRV board from the CN2 connectors on the infeed conveyor motor driver board (IFC)/ the weigh conveyor motor driver board (WTC) located at the lower part of the electrical control unit



Drive board (U4 [WTC])

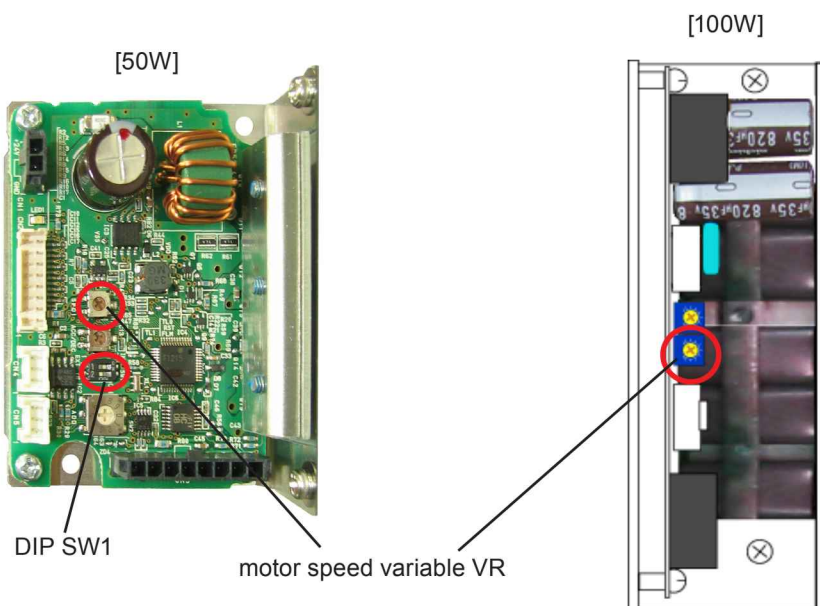
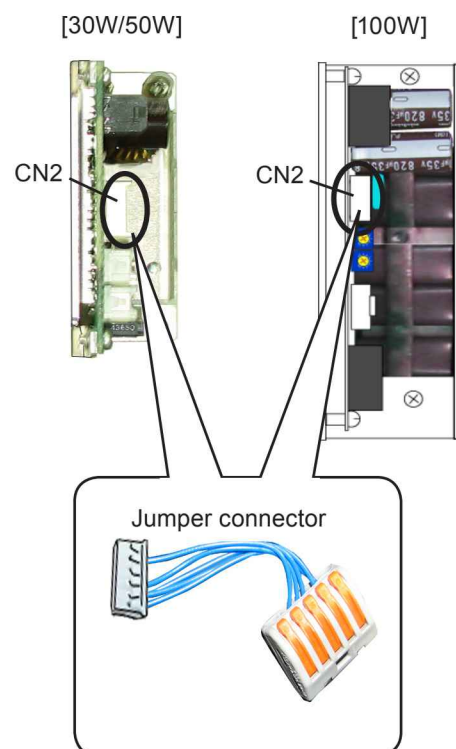


Drive board (U3 [IFC])



4. Insert jumper connectors attached to the cables (WTC-CN2, IFC-CN2) on the DRV board to the CN2 connectors on the infeed conveyor motor driver board (IFC)/the weigh conveyor motor driver board (WTC).
5. In this condition, turn the power switch ON, conduct the zero adjustment, and press the operation switch.
6. If the infeed/weigh conveyor belts do not start to move, turn the VR (motor speed variable VR) on the infeed/weigh conveyor motor driver boards in a clockwise direction. If the motor rotates, use the conveyors at the same speed used in the normal situation.

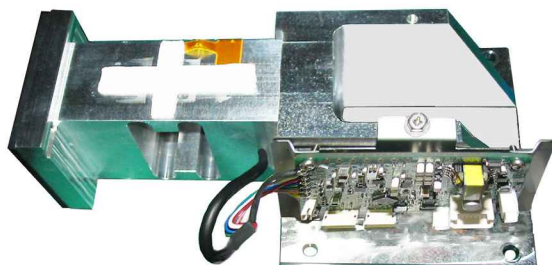
* For the 30 W/50 W motor driver, be sure to turn ON the DIP switch SW1 on the board.



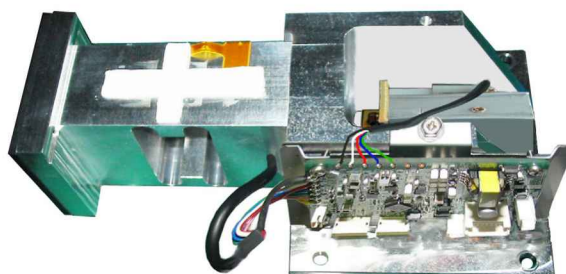
NOTE After the problem in the weigh unit is solved, do not forget to remove the jumper connector and use the machine over the regular connection.

3.10 Weigh Unit

■ Outline view of the unit



[Standard Sensor Unit]

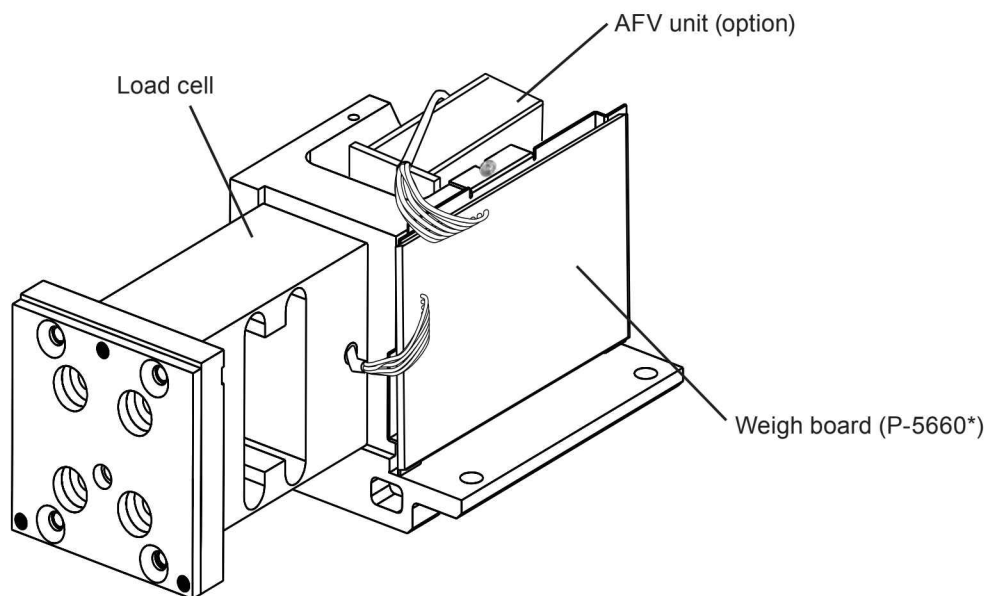


[Sensor Unit with AFV]

■ Unit Functions

- Weigh the product and send the A/D processed weigh signal to the DRV board via the weigher
- Send the photo eye unit signal to the DRV board via the weigher board.

■ Unit Detail View



NOTE

- Since the load cell and the weigh board are adjusted in a set (temperature correction/ four corners), replace them by the sensor unit.
- It is not necessary to adjust four corners after reassembly, yet check should be needed.

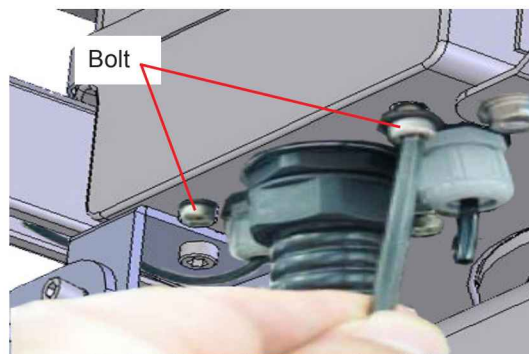
3.10.1 Replacement of Sensor Unit and Limitation Adjustment

Follow the procedures below as to replacement of the sensor unit and limitation adjustment after the replacement.

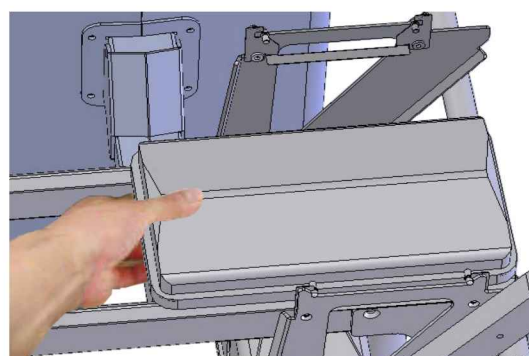


Here describes the operation after removing the drive box which is described at "3.8 Weigh/Drive Unit (SS Type)" and "3.9 Weigh/Drive Unit (CR Type)".

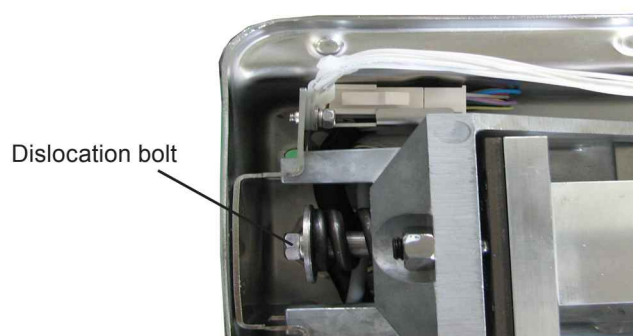
1. Remove the 2 bolts fixing the weigh box cover.



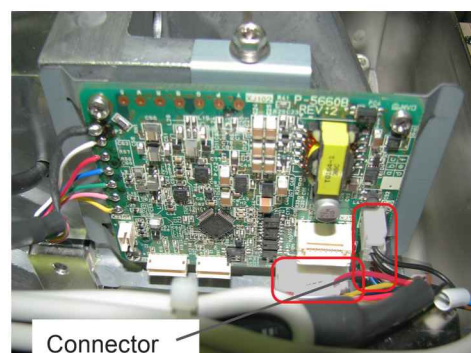
2. Lift the weigh box cover on the infeed side and remove it by sliding it slightly to the rejector side.



3. Remove the dislocation bolt attached to the sensor unit on the infeed side.



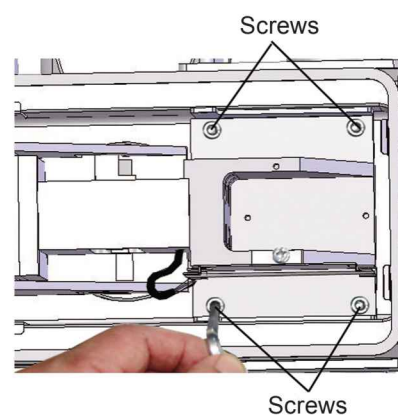
4. Remove the connectors of each lead wire (XJ4, XJ5, XJ6) from the weigher board.



3 Maintenance Adjustment and Replacement

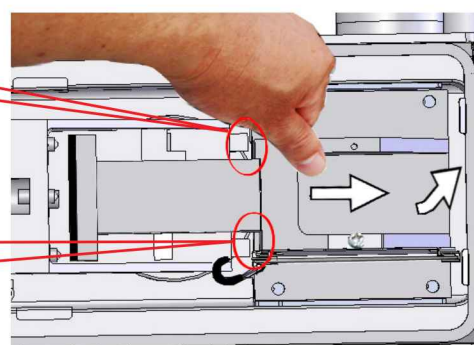
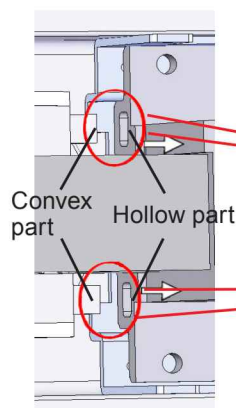
5. Remove the 4 screws fixing the sensor unit and remove the sensor unit.

- You can lift the sensor unit by detaching the convex part (base side) from the hollow part located at the lower part of the sensor unit.



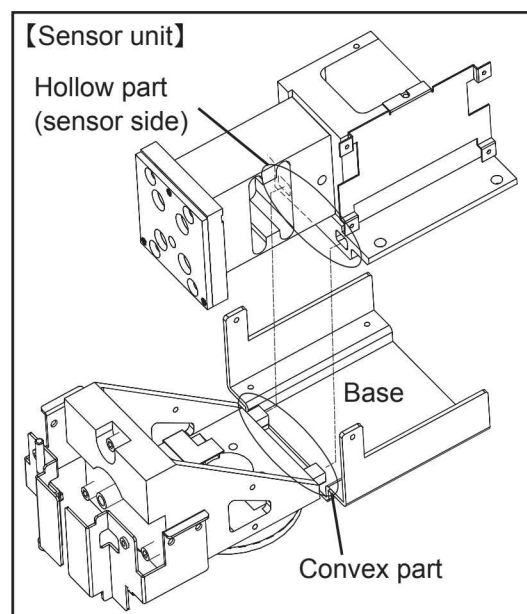
NOTE

When loosening the fixing screw, be careful not to apply excessive force to the sensor unit.



6. Slide the sensor unit to the rejector side and remove it by lifting it up.

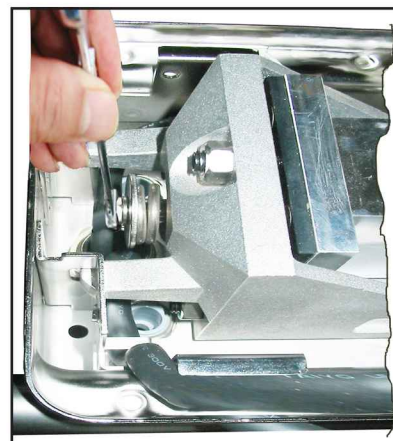
7. Replace the sensor unit.
Put the new sensor unit on the case and slide it to the infeed side. Make sure that the convex part (base side) and the hollow part located at the lower part of the sensor unit fit tightly.



NOTE

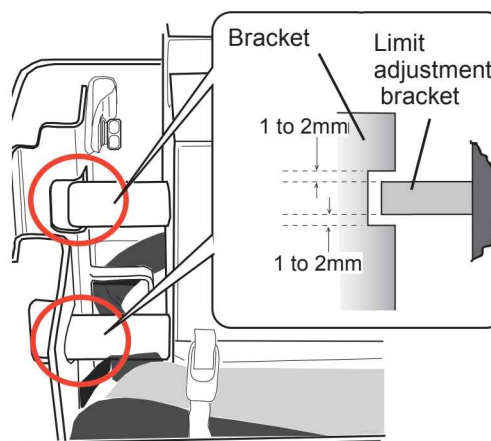
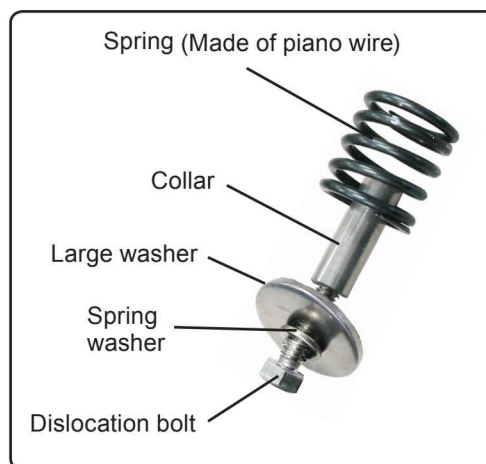
- Since the load cell and the weigher board are adjusted (temperature correction/ four corners) in a set, replace them by the sensor unit.
- It is not necessary to adjust four corners after reassembling. Just check that these four corners are fit tightly.
- When sending back the old sensor unit, put it in the designated case which was sent on replacement.

8. Assemble the sensor unit using 4 screws.
9. Connect the connectors of each lead wire (XJ4, XJ5, XJ6).
10. Attach the dislocation bolt.



Put the spring, the collar, the washers through the dislocation bolt (with nut), and attach to the cell base. Screw in the bolt approximately 10 mm, and tighten with the nut.

11. Check that a gap between the limit adjustment bracket and the cutout of the bracket on the base is 1 to 2 mm.
12. Put the weigh box cover and push it lightly on the rejector side to attach it to the weigh box.
13. Fix the weigh box cover by bolts from underneath of the weigh box.



It clicks when the cover is attached tightly.



REFERENCE

For attaching the optional AFV unit, refer to "4 ELECTRICAL UNIT CONFIGURATION".

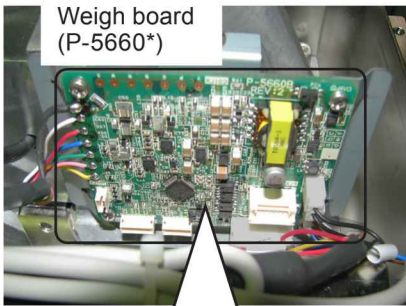
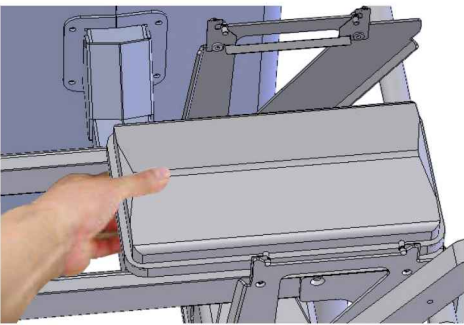
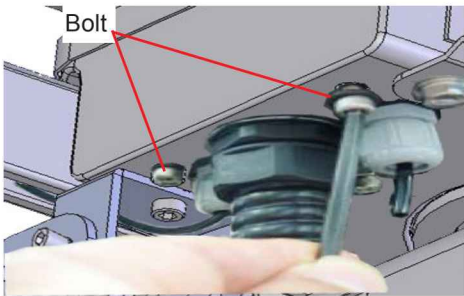
3.10.2 Judgment of Load Cell Breakage

To judge whether the load cell is broken or not, measure the output voltage of load cell. Measure the voltage output by the following steps:

- CAUTION**
- Even if the load cell is correct and only the weigh board is incorrect, replace them in a set.
 - The digital voltmeter used in the work here should be precise enough to allow the minimum display is approximately [0.0 mV].

■ Judging attached weigh conveyor and load cell

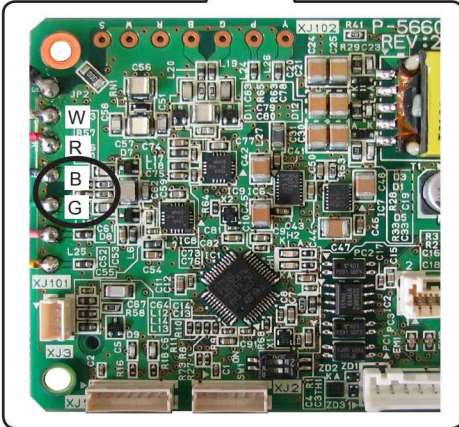
1. Remove the 2 bolts fixing the weigh box conveyor.
2. Remove the option parts (transfer plate and guide) mounted to the weigh conveyor.
3. Temporarily remove the weigh conveyor.
4. Lift the weigh box cover on the infeed side and remove it by sliding it slightly to the rejector side.
5. Connect the lead rod at the plus (+) side of digital voltmeter to [G], and connect the lead rod at the minus (GND) side to [B] of the weigh board (XJ101).
6. Attach the weigh conveyor.
7. Turn the power ON, and measure the voltage of the digital voltmeter to judge if the load cell is broken or not.



■ Criteria

- QLC-12L(DACS-GN-S015): -24.7 to 22.1 mV
- QLC-60L(DACS-GN-S060): -16.1 to 18.9 mV

If the measured voltage is within the above values, it is judged as normal.



■ Judging without load cell

1. Perform the step described in “3.10.1 Replacement of Sensor Unit and Limitation Adjustment” to remove the load cell and weigh board (P-5660*).
2. Remove the metals such as load cell base, and vertically stand only the load cell (QLC-12L/60L) as in the right picture to allow unloaded condition
3. Connect the load cell power connector (XJ5) coming from the instrumental unit to the connector XJ5 of weigh board.
4. Turn the power ON, and measure the differential voltage of the bridge circuit of load cell by using the digital voltmeter.
5. Connect the lead rod at the plus (+) side of digital voltmeter to [R] of the cell cable (red), and connect the lead rod at the minus (GND) side to [W] of the cell cable (white), and measure the voltage of the weigh board (XJ101).

Rated voltage: 27 V

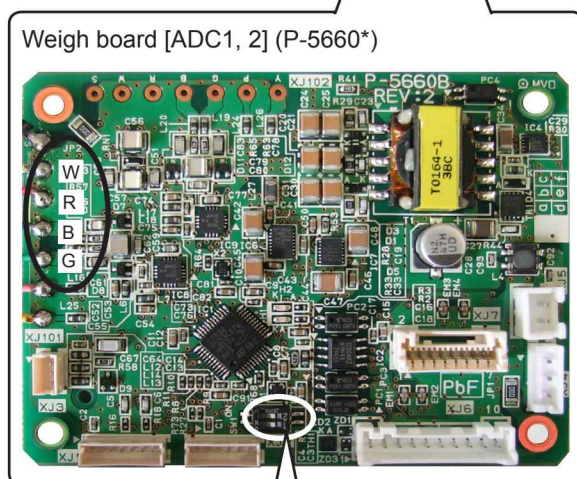
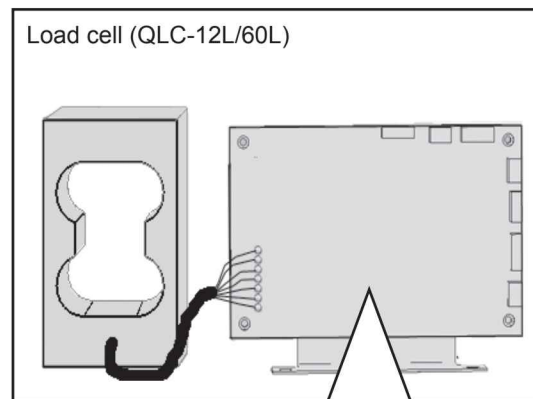
6. Connect the lead rod at the plus (+) side of digital voltmeter to [G] of the cell cable (green), and connect the minus (GND) side to [B] of the cell cable (blue), and measure the voltage of the weigh board (XJ101).

■ Criteria

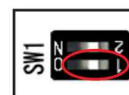
The initial bridge balance voltage of the load cell is as follows (when the applied voltage is 27 V);

- QLC-12L(DACS-GN-S015): -49.2 to -35.8 mV
- QLC-60L(DACS-GN-S060): -21.0 to 1.0 mV

If the measured voltage is within the above values, it is judged as normal.



SW1-1 on Weigh board [ADC 2] (P-5660*): OFF



Weigh board [ADC2] (Weigh side)

SW1-1 on Weigh board [ADC 1] (P-5660*): OFF



Weigh board [ADC1] (Infeed side)
(option)

